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Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools

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Contents

Abstract

Mass spectrometry is a cornerstone technique in analytical chemistry, metabolomics, pharmaceutical research, and many other fields. The prediction of mass spectra from molecular structures, and the inverse inference of molecular structures from spectra, remain active areas of research. While machine learning methods increasingly improve forward spectrum prediction from known structures, the inverse problem — predicting plausible molecular structures from a given spectrum — remains difficult, ambiguous, and poorly interpretable.

This work investigates the integration of auxiliary intermediate fragments generated by graph-transformation tools into machine learning pipelines for bidirectional translation between molecular structures and mass spectra. These graph-based intermediate representations provide an explicit fragmentation layer between molecules and spectra, intended to make the prediction process more transparent than purely black-box approaches.

The proposed framework systematically incorporates graph-based fragmentation intermediates into structure $\leftrightarrow$ spectrum translation models. By combining molecular representations, spectral information, and fragment-level graph structures, it establishes a functional bidirectional pipeline with an explicit intermediate fragmentation layer. In a proof-of-concept evaluation, a forward-model reranking stage substantially improves retrieval, whereas the fragment-aware components do not yet yield a measurable gain over ablated variants under the small-sample setting. The principal contribution is therefore the framework itself, together with a systematic analysis of its strengths, limitations, and failure modes, providing a concrete basis for future work on learned fragment generation.

Zusammenfassung

Massenspektrometrie ist eine Schlüsseltechnik in der analytischen Chemie, der Metabolomik, der pharmazeutischen Forschung und vielen weiteren Bereichen. Die Vorhersage von Massenspektren aus Molekülstrukturen sowie die inverse Ableitung von Molekülstrukturen aus Spektren bleiben aktive Forschungsbereiche. Während Methoden des maschinellen Lernens die Vorhersage von Spektren aus bekannten Strukturen zunehmend verbessern, bleibt das inverse Problem — die Vorhersage plausibler Molekülstrukturen aus einem gegebenen Spektrum — schwierig, mehrdeutig und nur eingeschränkt interpretierbar.

Diese Arbeit untersucht die Integration zusätzlicher Zwischenfragmente, die durch graphbasierte Transformationstools erzeugt werden, in Machine-Learning-Pipelines für die bidirektionale Übersetzung zwischen Molekülstrukturen und Massenspektren. Diese graphbasierten Zwischenrepräsentationen bilden eine explizite Fragmentierungsschicht zwischen Molekülen und Spektren und machen den Vorhersageprozess dadurch besser interpretierbar als rein Black-Box-basierte Ansätze.

Das vorgeschlagene Framework integriert graphbasierte Fragmentierungszwischenstufen systematisch in Modelle zur Struktur $\leftrightarrow$ Spektrum-Übersetzung. Durch die Kombination molekularer Repräsentationen, spektraler Information und fragmentbasierter Graphstrukturen entsteht eine funktionsfähige bidirektionale Pipeline, die durch ihre Konstruktion interpretierbar ist. In einer Machbarkeitsstudie (Proof of Concept) verbessert eine auf dem Vorwärtsmodell basierende Reranking-Stufe die Wiederauffindung (Retrieval) deutlich, während die fragmentbasierten Komponenten unter den gegebenen Kleinstichprobenbedingungen noch keinen messbaren Vorteil gegenüber Ablationsvarianten erbringen. Der wesentliche Beitrag besteht daher im Framework selbst sowie in einer systematischen Analyse seiner Stärken, Grenzen und Fehlermodi, die eine konkrete Grundlage für zukünftige Arbeiten zur gelernten Fragmenterzeugung bildet.

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Chapter 1

Introduction

1.1 Motivation

Mass spectrometry (MS) is one of the most widely used analytical techniques for molecular characterization across chemistry, metabolomics, environmental sciences, and pharmaceutical research. [1, p. 1] Its high sensitivity [2, p. 1764], reproducibility [2, p. 1764], and scalability [2, p. 1764] have enabled large-scale molecular profiling in both research and industrial settings. Modern instrumentation such as Orbitrap, Q-TOF, and triple quadrupole mass spectrometers generates high-throughput spectral data that can be compared against large repositories of molecular fingerprints such as the NIST Mass Spectral Library to support compound identification and profiling [3, pp. 7274-7275].

Despite these advances in data acquisition, the interpretation of MS data remains a significant bottleneck [4, p. 1]. Structural elucidation from spectra is still largely expert-driven, relying on chemical intuition and established fragmentation rules [4, p. 11]. While thousands of spectra can be recorded within a short time frame [4, p. 1], the corresponding structural annotation process remains comparatively slow, manual, and difficult to scale [4, p. 3]. This imbalance creates a fundamental asymmetry between data generation and data analysis.

The intrinsic complexity of molecular fragmentation contributes substantially to this challenge [1, p. 240]. Fragmentation processes are governed by chemically valid bond cleavages and rearrangements [1, pp. 30-31], leading to multiple competing fragmentation pathways. A single molecule may therefore generate numerous fragments through direct cleavages and rearrangement reactions [1, p. 240-241]. Because fragmentation pathways depend sensitively on molecular structure and internal energy [1, p. 26, pp. 30-31], structurally distinct molecules may produce partially overlapping spectral signatures, while small structural variations can induce significant spectral differences. Consequently, the inverse problem — inferring molecular structure from spectral data — is inherently ill-posed [1, p. 240].

Recent machine learning approaches aim to address this challenge by learning direct mappings between molecular representations and spectral outputs. Graph neural networks [5], sequence-based encoders [6, p. 1573], and generative models have demonstrated promising performance in forward (molecule-to-spectrum) and inverse (spectrum-to-molecule) prediction tasks. However, purely data-driven approaches exhibit several limitations:

- They often lack mechanistic interpretability. [7, p. 3& 6]
- They do not explicitly model fragmentation pathways. [7, p. 3]
- They rely on statistical correlations rather than chemically structured priors. [8, p. 966]
- Their generalization to rare or out-of-distribution molecular scaffolds remains limited. [9, p. 12]

In particular, models that treat molecules as flat graphs or string encodings and spectra as unordered peak lists ignore the structured chemical processes underlying fragmentation. This omission reduces interpretability and may hinder robust extrapolation beyond the training distribution. [10, p. 3420]

Fragmentation in mass spectrometry is not an arbitrary stochastic process; it follows chemically constrained transformation sequences. [11, ch. 4& 8] Therefore, incorporating structured representations of fragmentation—such as derivation graphs encoding valid transformation rules—offers a principled way to introduce chemical inductive bias into machine learning models. By explicitly modeling fragment lineage and transformation pathways, it becomes possible to align molecular substructures with spectral evidence in a mechanistically meaningful manner.

This thesis is motivated by the hypothesis that integrating chemically structured derivation graphs into a machine learning framework can introduce chemical inductive bias and yield an explicit fragmentation layer, while supporting bidirectional structure-spectrum modeling. By combining fragment-aware representations with data-driven learning, the aim is to bridge the gap between high-throughput spectral acquisition and scalable, chemically grounded interpretation.

1.2 Objectives

The objective of this thesis is to design, implement, and systematically evaluate a fragment-augmented machine learning framework for improving the bidirectional translation between molecular structures and mass spectra. Specifically, the work investigates whether the integration of graph-based reaction and fragment derivation graphs, generated through chemically valid graph transformation rules, can serve as structured intermediate representations, and to what extent they affect predictive performance.

Mass spectrometry-based structure elucidation represents an inherently ill-posed inverse problem: structurally distinct molecules may yield partially overlapping spectral signatures, while minor structural modifications can induce substantial spectral changes. This thesis therefore introduces fragment-level intermediates derived from rule-based graph transformations to inject chemically grounded structure into the learning process.

Instead of directly learning a mapping between molecular graphs and spectral representations, the proposed framework decomposes the task into a structured pipeline in which molecular graphs are transformed into chemically plausible fragment sets, which in turn mediate the relationship to spectral outputs. These fragment sets are embedded into a shared latent space together with molecular and spectral representations, enabling aligned embeddings across the domains.

Ultimately, this thesis assesses to what extent incorporating chemically valid graph-based fragment derivations into neural learning pipelines affects the predictive power of bidirectional structure-spectrum modeling in mass spectrometry, and outlines how such a framework could enable de novo molecular generation from spectral data.

1.3 Scope and Limitations

This thesis is restricted to electron ionization (EI) mass spectrometry, as EI fragmentation reactions are comparatively well understood and mechanistically characterized. [11, ch. 4& 8] This allows the use of chemically informed rule-based graph transformations to generate fragment derivation graphs under consistent assumptions. Other ionization techniques are excluded due to their differing fragmentation behavior and the additional modeling complexity they would introduce.

A major limitation arises from the computational cost of graph-based fragment generation. Rule-based graph transformations lead to combinatorial growth in possible fragments, [12, p. 4] especially when multiple reaction rules and recursive fragmentation steps are considered. Ensuring chemical validity and avoiding duplicate graph states further increases computational demands. [13, pp. 2-3]

Due to this scalability constraint, the experimental evaluation is limited to order of magnitude of 100 compounds. While sufficient for demonstrating methodological feasibility, this restricted dataset limits chemical diversity and does not allow for generalization. Consequently, the results should be interpreted as proof-of-concept validation rather than evidence of large-scale applicability.

It is also important to note that the reactions are defined in terms of graph symbols in an ambiguous manner. Furthermore, the transformation rules are not exhaustive of all possible fragmentation pathways, and some may even be incorrect. This means that the generated fragment derivation graphs may not capture all relevant chemical transformations, and the model's performance may be sensitive to the specific choice of transformation rules. Therefore, the generated fragment derivations are constrained by the assumptions embedded in the rule set. Consequently, it may not generalize to fragmentation processes that deviate from these assumptions.

Chapter 2

Background and Related Work

2.1 Fundamentals and Capabilities of Mass Spectrometry

Mass spectrometry is a core analytical technique that characterizes chemical species by converting them into gas-phase ions and measuring their mass-to-charge ratio (m/z). [14, p. 5] A mass spectrometer therefore acts as a highly sensitive "molecular balance" whose output – a mass spectrum – reports which ionic species are present and with what relative abundances.

Conceptually, an MS experiment comprises three main steps:

(i) **ionization**, in which neutral molecules are transferred into the gas phase and charged; [14, p. 1]

(ii) **mass analysis**, in which ions are separated according to m/z; and

(iii) **detection**, in which ion signals are recorded and assembled into a spectrum. [14, p. 5]

Because many compounds form multiple ion species (e.g., adducts [14, p. 21], isotopologues [1, p. 67], fragments), the measured spectrum is not a direct image of the neutral molecule but a structured measurement that must be interpreted in the context of the ionization and fragmentation processes used. [14, p. 16-18]

A key capability of MS is the *identification of unknown compounds*. Even without prior knowledge of the analyte, MS provides constraints that narrow the space of possible identities. Accurate m/z measurements can restrict candidate molecular formulas; isotope patterns (e.g., from ^{13}C, ^{37}Cl, ^{81}Br) further refine these candidates by providing diagnostic abundance ratios and characteristic spacings. [1, p. 263] When fragmentation data are available – either through tandem MS (MS/MS) or through ionization methods that induce fragmentation – structural hypotheses can be tested by matching observed fragment ions and neutral losses to expected substructures. [1, p. 225] In practice, unknown identification often combines these orthogonal cues with computational searches in spectral libraries and structure databases, where a measured spectrum is matched against reference spectra or predicted fragmentation patterns. [15, p. 8] For complex samples, chromatographic coupling (GC-MS or LC-MS) adds retention time as an additional dimension, [1, p. 475] improving selectivity and reducing confounding effects from co-eluting compounds. [14, p. 233-237]

Also in *purity control and quality assurance* MS serves a critical role, particularly in pharmaceutical manufacturing. By detecting the presence and abundance of impurities, degradation products, and unexpected byproducts, MS ensures that manufactured compounds meet specification and quality standards. The high selectivity and sensitivity of MS make it ideal for identifying trace contaminants that would escape less discriminating analytical techniques. Combined with chromatographic separation, MS provides definitive identification and quantification of both the target compound and potential impurities in a single analytical run. [16, p. 2]

MS is equally important for the *quantification of known substances*, where the goal is to estimate analyte concentration rather than molecular identity. [14, p. 260] Quantitative MS is grounded in the relationship between the number of detected ions and the amount of analyte introduced into the instrument. [14, p. 261 & 265-266] In idealized settings, ion signal intensity scales proportionally with concentration; in real measurements, this proportionality is skewed. Consequently, reliable quantification typically relies on calibration strategies. *External calibration* constructs a response curve from standards of known concentration, while *internal standardization* – often using isotopically labeled analogs – compensates for variability during sample preparation, injection, and ionization. [14, p. 264-268] The outcome is a method that can measure trace concentrations across wide dynamic ranges, which is central in many applications.

Beyond identity and amount, MS provides a systematic route to *deduce structural features of molecules*. Structural inference is primarily enabled by fragmentation behavior: when a precursor ion breaks apart, the resulting fragment ions reflect chemically plausible bond cleavages and rearrangements, thereby encoding information about functional groups, substructures, and connectivity. [17, p. i49] Certain fragmentation pathways are characteristic – for example, losses of small neutrals such as H_2O, NH_3, or CO can indicate the presence of hydroxyl, amino, or carbonyl functionalities, while specific fragment series can suggest alkyl chain lengths or aromatic substitution patterns. [1, p. 504] Importantly, structural elucidation from MS data is rarely straight forward; rather, it is an inference problem in which multiple candidate structures may explain the same spectral evidence, especially for isomers. [18, p. 53]

As a result, modern workflows increasingly integrate MS measurements with computational models that generate and rank candidate structures based on spectral consistency.

2.2 Principles of Mass Spectrometry

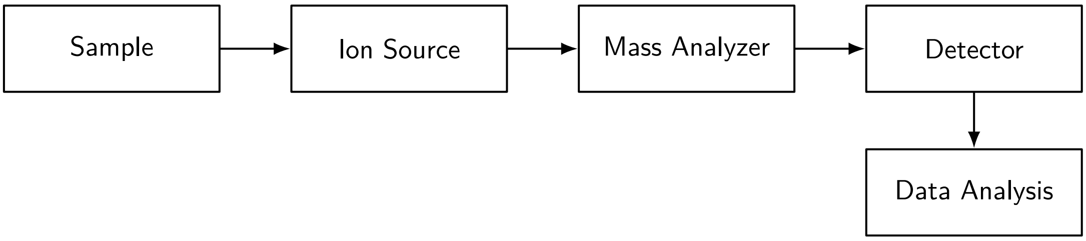


Figure 2.1: Simplified block diagram of a mass spectrometer. The sample is introduced into the ion source, where analyte molecules are ionized. The resulting ions are separated in the mass analyzer according to their mass-to-charge ratio (m/z), detected by the detector, and subsequently processed in the data analysis step. [19]

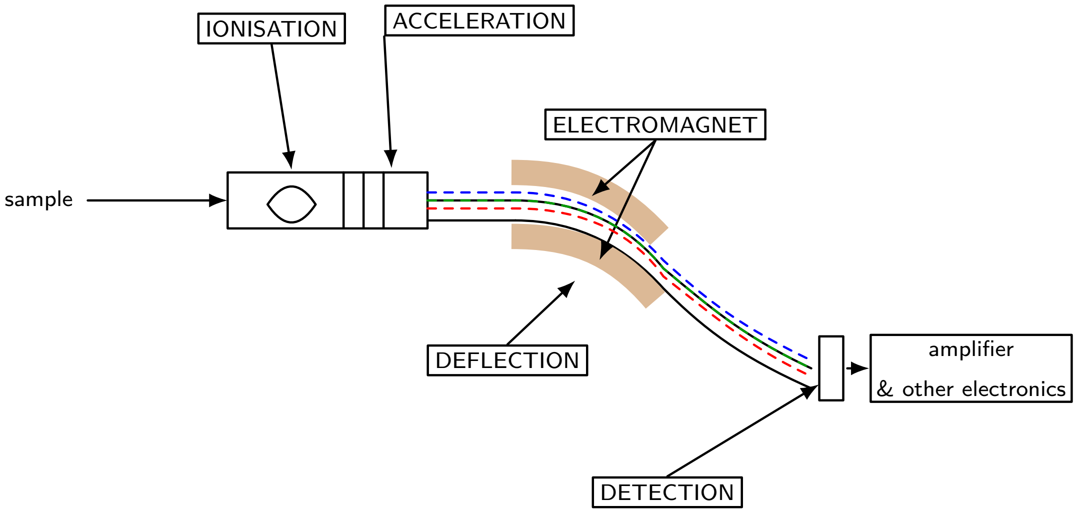


Figure 2.2: Simplified schematic of the measurement principle in a magnetic sector mass spectrometer. Following sample introduction, ionisation and acceleration generate an ion beam that is separated in the magnetic field by deflection that depends on m/z. The separated ions are then detected and transformed into an electrical signal for further processing. [20]

Instrument workflow

Figure 2.1 shows steps in a mass spectrometer: a *sample* is introduced, converted into gas-phase ions in the ion source, separated according to motion in the *mass analyzer* that depends on mass, detected in the *detector*, and finally interpreted during *data analysis*. The central observable is the mass to charge ratio m/z. A mass spectrum is therefore a distribution of signal intensity as a function of m/z, where the signal reflects the abundance of ions reaching the detector. [21, p. 1566] In most practical contexts, intensities are reported as relative intensity, normalized such that the most intense peak equals 100% of the base peak. [21, p. 1550] The x-axis is conventionally labeled with m/z (mass to charge ratio), a dimensionless quantity. Labeling with mass units such as *daltons* (Da), amu, or unified atomic mass unit (u) is discouraged – although common in practice – as it causes confusion with multiple charged ions. [21, p. 1516]

Ion notation and precursor concepts

Ions observed in mass spectrometry are typically derived from a neutrally charged molecule M. In electron ionization (EI) a hard ionization method, removal of an electron yields the molecular ion [21, p. 1541] – also called the parent ion – denoted $[M]^{+\cdot}$, [21, p. 1582] a radical cation carrying both a positive charge and an unpaired electron. More generally, the term *precursor ion* refers to the ion selected for fragmentation, [21, p. 1578] from which product ions are generated. In soft ionization methods like eloctrospray, ion formation is often dominated by proton transfer, producing a protonated molecule $[M + H]^+$. [21, p. 1580]

Peak types: isotope, fragment, and base peaks

A measured spectrum contains several conceptually distinct peak types. *Isotope peaks* arise from naturally occurring heavier isotopes (e.g., ^{13}C, ^{37}Cl), producing characteristic $M + 1, M + 2, \ldots$ patterns adjacent to the monoisotopic ion peak. *Fragment peaks* arise when precursor ions undergo dissociation, yielding smaller product ions whose m/z values and relative intensities encode structural information. The *base peak* is the most intense peak in the spectrum [21, p. 1524] and is used for normalization; [21, p. 1550] it is not necessarily the molecular ion and often corresponds to a particularly stable fragment. [22, p. 2] [1, p. 263]

2.2.1 Ionization and Fragmentation

Ionization techniques are broadly classified into two categories based on the energy imparted to the analyte: *hard ionization* methods, which deliver high internal energy and promote extensive fragmentation, [21, p. 1547] and *soft ionization* methods, which minimize fragmentation and favor molecular ion preservation. [21, p. 1591] [23, p. 34]

Hard Ionization methods

Electron Ionization (EI) is a hard ionization technique in which a stream of high-energy electrons (typically 70 eV) collides with neutral molecules in the gas phase. [1, p. 197] The collision transfers energy to the molecule, causing the ejection of an electron and formation of a radical cation $[M^{+\cdot}]$ with both a positive charge and an unpaired electron. [1, p. 194] Because the internal energy of the molecular ion typically exceeds the activation barriers for bond cleavage, the $[M^{+\cdot}]$ ion spontaneously fragments into smaller product ions and neutral molecules. [24, p. 3] As a result, EI mass spectra are characterized by rich fragmentation patterns that reflect the molecular structure. [25]

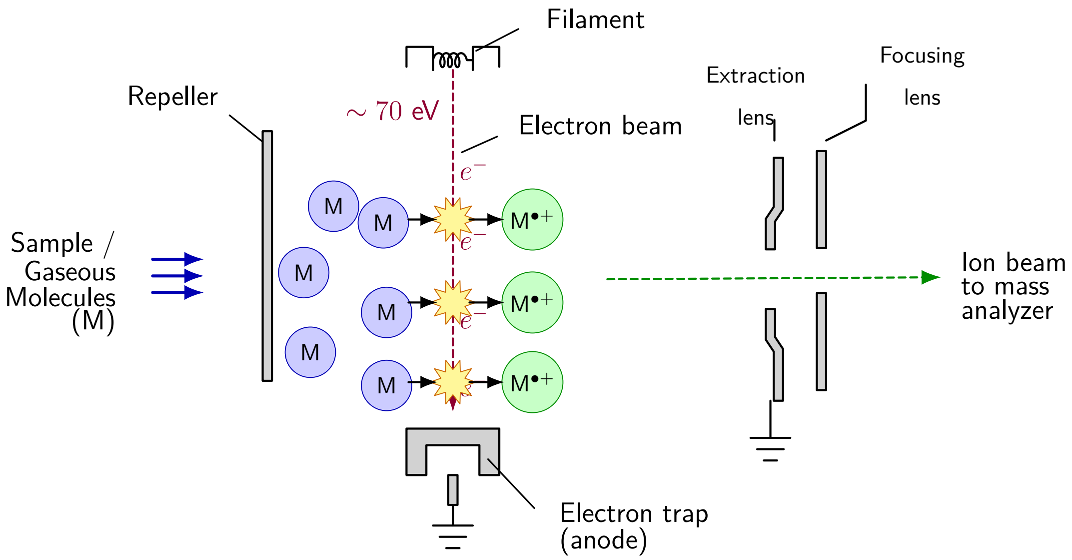


Figure 2.3: Schematic illustration of the electron ionization (EI) process. Gaseous sample molecules (M) enter the source and are crossed by a beam of high-energy electrons ($\sim 70\,\text{eV}$) emitted from a heated filament and collected at the electron trap (anode). Collision with the electron beam ejects an electron from each molecule, forming a radical cation $M^{\bullet+}$. A repeller pushes the resulting ions toward the extraction and focusing lenses, which shape them into an ion beam directed to the mass analyzer. [26, p.18]

The reproducible nature of EI fragmentation pathways makes electron ionization particularly valuable for structural elucidation. [27] Characteristic losses of small neutral species provide diagnostic clues about functional groups and connectivity. [28] EI is the dominant ionization method in gas chromatography-mass spectrometry (GC-MS) and remains widely used in small-molecule analysis. [29, p. 1]

Spark Ionization (SI) , is a hard ionization technique [30, p. 1], that often utilized to detect trace elements at parts-per-billion (ppb) levels. [31, p. 428] It uses high-voltage up to 100 kV pulsed electric sparks between two electrodes to vaporize, atomize, and ionize conductive or semi-conductive materials. [31, p. 428]

Soft Ionization methods

Electrospray Ionization (ESI) is a prominent soft ionization technique particularly well-suited for coupling with liquid chromatography (LC-MS) and for analysis of large biomolecules. [32] In ESI, a liquid sample is converted into a fine aerosol of charged droplets by applying a high electric potential (typically several kilovolts) between a capillary needle and a counter electrode. [33, p. 3] As the droplets travel toward the mass analyzer, they undergo solvent evaporation and coulombic fission, eventually releasing gas-phase ions into the vacuum of the spectrometer. [33, p. 7] [26, p. 53]

ESI ionization is exceptionally gentle, with minimal or no fragmentation in the ion source. [33, p. 7] This characteristic makes ESI ideal for determining accurate molecular weights of intact molecules. [26, p. 211] Notably, ESI can generate multiply charged ions $[M + nH]^{n+}$ from large molecules, reducing their effective m/z and enabling their analysis on instruments with limited mass range. [26, p. 50]

A curtain gas is often employed to shield the ion path, preventing neutral contaminants and solvent droplets from reaching the mass analyzer while allowing ions to pass through. [14, p. 42] The sequence from charged droplet formation to gas-phase ion release is shown in Figure 2.4.

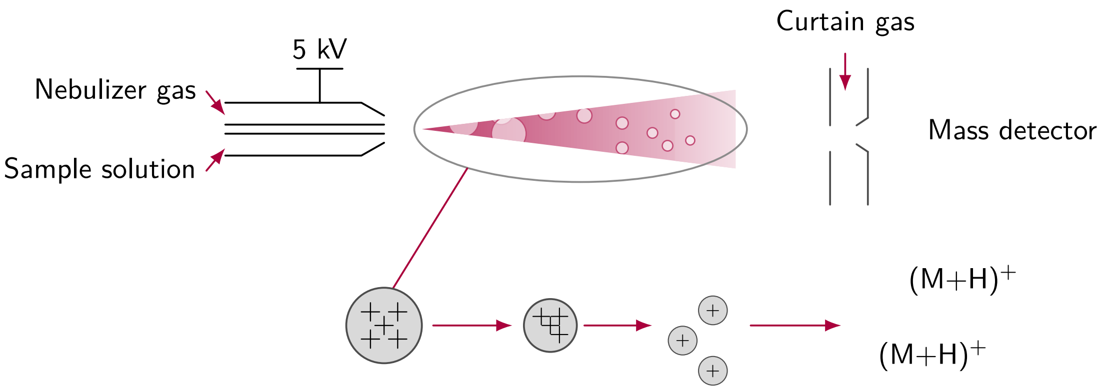


Figure 2.4: Schematic illustration of the electrospray ionization (ESI) process. A sample solution is dispersed by nebulizer gas and subjected to a high electric potential, producing a spray of charged droplets. Progressive solvent evaporation and droplet fission lead to the formation of gas-phase ions, which are transferred through the interface region into the mass detector. The figure highlights the transition from charged droplets to molecular ions, exemplified here by protonated species $(\mathrm{M} + \mathrm{H})^+$.

[23, p.39]

Atmospheric Pressure Chemical Ionization (APCI) is a soft ionization method used primarily in LC-MS, [23, p. 39] where analytes are ionized at atmospheric pressure via ion-molecule chemistry initiated by a corona discharge. [21, p. 1522] In contrast to ESI, no high voltage is applied to the inlet capillary; instead, the mobile phase is nebulized, rapidly evaporated in a heated region, [26, p. 56] and the resulting vapor passes a high-voltage needle that generates reagent ions. Solvents dominate the early ionization steps [23, p. 39] and effectively act as a reagent gas, transferring charge to neutral analytes mainly by proton transfer to form $[M+H]^+$.

Because APCI imparts relatively little internal energy, spectra typically show limited fragmentation and are well suited to quantitative LC-MS/MS, [26, p. 171] where signal is concentrated into a few precursor-ion peaks. A heated transfer tube and/or countercurrent curtain gas is used to decluster solvent-analyte adducts prior to ion extraction. [23, p. 39] APCI is often more efficient than ESI for less polar compounds, [1, p. 466] but its higher source temperatures can induce thermal degradation. [23, p. 39]

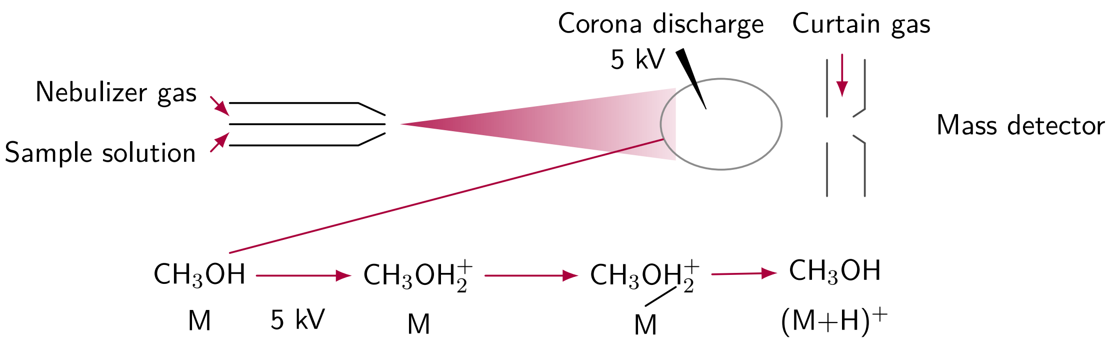


Figure 2.5: Schematic illustration of atmospheric pressure chemical ionization (APCI). The sample solution is nebulized and vaporized, and a corona discharge operated at high voltage generates reagent ions from the solvent. In the example shown, protonated ions transfer a proton to the analyte molecule M, forming the protonated molecular ion $(M+H)^+$, which is then introduced through the interface region into the mass detector.

[23, p.39]

MALDI (Matrix-Assisted Laser Desorption/Ionization) [34] is a soft ionization technique [35]. that uses a laser to desorb and ionize analytes from a solid matrix. [26, p. 36] It is particularly effective for large biomolecules and polymers, [14, p. 33][26, p. 36] producing predominantly singly charged ions. [14, p. 35]. Tuning laser energy and matrix composition can optimize ionization efficiency and control fragmentation. A general schematic of the MALDI process is shown in Figure 2.6.

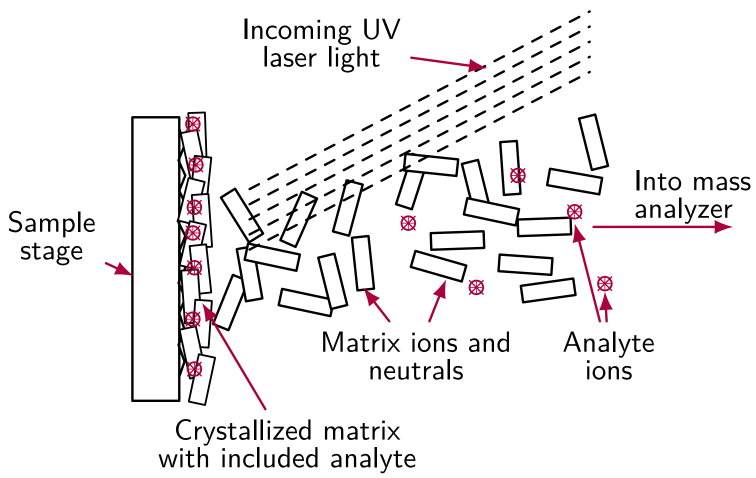


Figure 2.6: Schematic illustration of matrix-assisted laser desorption/ionization (MALDI). A crystallized matrix containing embedded analyte molecules is irradiated with UV laser light at the sample stage, leading to desorption and ionization of matrix and analyte species. During plume formation, matrix ions and neutrals promote the formation of analyte ions, which are subsequently transferred into the mass analyzer.

[23, p. 42]

2.2.2 Mass Analyzers

After ionization, the generated ions must be separated according to their mass-to-charge ratio m/z before detection. This separation is performed by the *mass analyzer*. [26, pp. 5-6] Different analyzer designs exploit electric and magnetic fields to distinguish ions with different m/z values. Among the most widely used analyzers are magnetic sector instruments, time-of-flight (TOF) analyzers, and quadrupole analyzers. [26, pp. 108-109]

Magnetic Sector Mass Analyzers

Magnetic sector analyzers separate ions according to their motion in a magnetic field. The motion of a charged particle with mass m and charge q in electric and magnetic fields is described by the Lorentz force equation

$$\left(\frac{m}{q}\right)\mathbf{a} = \mathbf{E} + \mathbf{v} \times \mathbf{B},$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ the magnetic field, $\mathbf{v}$ the ion velocity, and $\mathbf{a}$ the acceleration. Ions with identical mass-to-charge ratios experience identical forces and therefore follow identical trajectories inside the analyzer.

When ions enter a magnetic field perpendicular to their velocity, the Lorentz force bends their trajectory into a circular path. The balance between the magnetic force and the centripetal force yields

$$qvB = \frac{mv^2}{r},$$

with r being the radius of the circular path, which leads to

$$\frac{m}{q} = \frac{B^2r^2}{2V},$$

assuming the ions were accelerated through a potential V. Ions with different m/z therefore follow trajectories with different radii and can be spatially separated. [36, pp. 1799-1800]

Time-of-Flight (TOF) Analyzers

Time-of-flight analyzers separate ions according to the time required to travel a fixed distance after acceleration. When ions are accelerated through a potential V, they acquire kinetic energy

$$\frac{1}{2}mv^2 = qV.$$

Solving for the velocity gives

$$v = \sqrt{\frac{2qV}{m}}.$$

In a field-free drift region of length L, the flight time is therefore

$$t = \frac{L}{v} = L\sqrt{\frac{m}{2qV}}.$$

Consequently, lighter ions reach the detector earlier than heavier ions, allowing determination of their m/z values. [37, p. 1152]

Quadrupole Mass Filter (QMF)

Quadrupole analyzers consist of four parallel rods to which a combination of radio-frequency (RF) and direct current (DC) voltages is applied. [38, p. 2] The resulting oscillating electric field produces ion trajectories that are stable only for ions within a specific range of mass-to-charge ratios. Ions with stable trajectories pass through the quadrupole to the detector, while ions with unstable trajectories collide with the rods and are removed. [39, p. 1] By scanning the applied voltages, ions of increasing m/z can be transmitted sequentially, generating a mass spectrum. [38, p. 2]

Quadrupole Ion Trap (QIT)

The quadrupole ion trap operates on the same fundamental principle as the linear quadrupole analyzer described above, added with the ability of oscillating radiofrequency fields to selectively confine ions according to their mass-to-charge ratio. [40, p. 1156] The key distinction lies in the geometry of the electric field: whereas the linear quadrupole uses four parallel rods to create a two-dimensional confining potential that allows ions to pass through along the axial direction, the ion trap employs a three-dimensional quadrupolar field configuration that confines ions in all spatial dimensions within a central cavity. [41, p. 962] This fundamental geometric difference enables the QIT to accumulate and store ions over time [41, p. 973], rather than merely filtering them during transit, which has profound implications for sensitivity, scan modes, and the ability to perform multiple stages of mass analysis within a single device. [41, p. 984]

2.2.3 Detection

After ions have been separated by the mass analyzer, they must be detected and converted into a measurable electrical signal. [26, pp. 5-6] The detector is transforming the arrival of individual ions into quantifiable data that can be assembled into a mass spectrum. [26, p. 103]

The dominant detector in Electron Ionization Mass Spectrometry is the Electron Multiplier (EM). [26, p. 103]

An electron multiplier operates by converting a single incoming ion into a large cascade of electrons through the following sequence:

1. An ion hits the first dynode
2. Secondary electrons are emitted
3. These electrons are accelerated to the next dynode
4. The process repeats (10-20 stages)
5. Result: a measurable current pulse

This cascade amplification principle is illustrated in Figure 2.7.

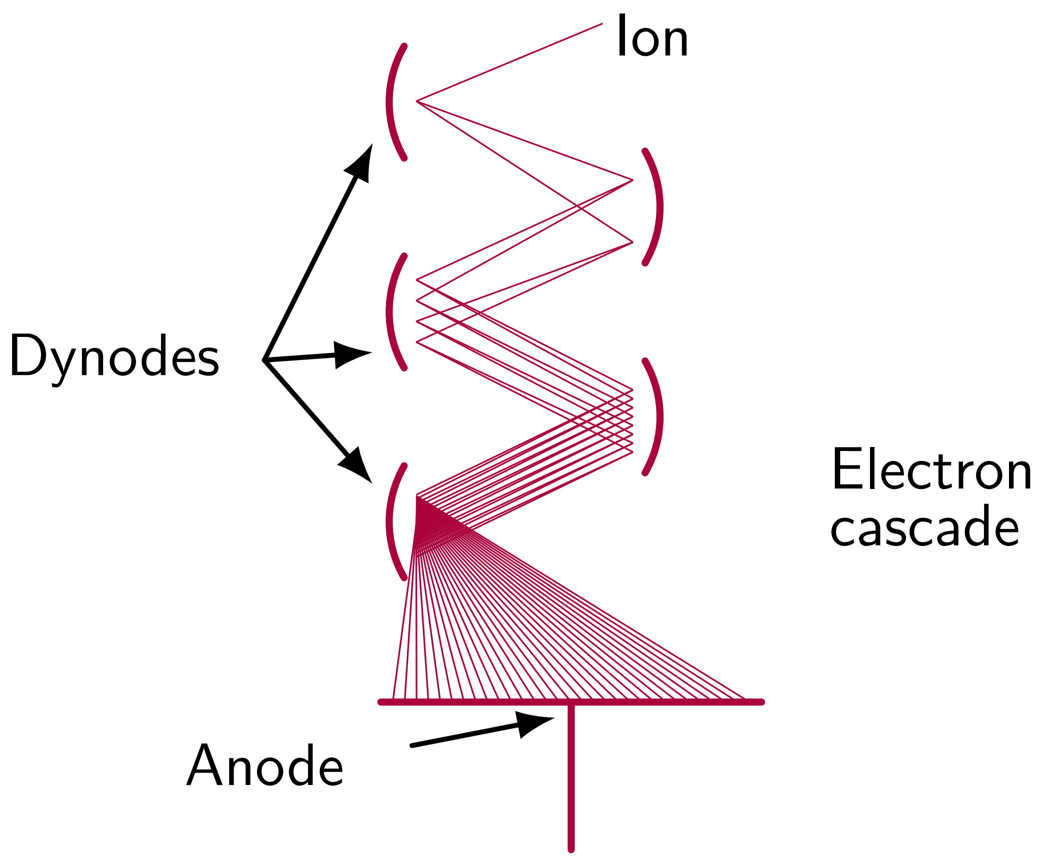


Figure 2.7: Schematic illustration of an electron multiplier (EM) detector. An incoming ion strikes the first dynode, causing the emission of secondary electrons. These electrons are then accelerated to subsequent dynodes, where they generate additional electrons in a cascading effect. After multiple stages, this results in a significant amplification of the original ion signal, producing a measurable current pulse that can be digitized and recorded as part of the mass spectrum.

[23, p. 50]

The amplification factor of an electron multiplier can reach 10^6 to 10^8, enabling the detection of individual ions with high sensitivity. The resulting current pulses are digitized and recorded as a function of m/z, producing the final mass spectrum that serves as the basis for subsequent data analysis. [26, p. 104]

2.2.4 Manual Interpretation of the MS Spectrum

Data analysis transforms the raw spectrum into structural hypotheses: isotope patterns constrain elemental composition, [1, p. 67] while fragment ions and neutral-loss relationships constrain functional groups and connectivity. [28] Note, multiple candidate structures can be compatible with the same spectral evidence, particularly for isomers under hard ionization methods like EI. [42, p. 6996]

The following summarize a typical manual workflow [26, pp. 215-216] for interpreting EI mass spectra.

1. Determine the molecular ion peak (M) and its isotopic pattern

2. find isotope patterns (e.g., Cl, Br, S)

Table 2.1: Sample Isotope Patterns

Element	Diagnostic pattern
Cl	3:1 doublet (M, M+2)
Br	1:1 doublet
S	small M+2 peak
C	^{13}C gives predictable M+1

3. determine the base peak (100%) which is the energetically favored fragmentation

4. analyze fragment differences and identify neutral losses

5. match fragmentation to chemical logic defined by reaction rules

These practical heuristics are grounded in specific EI reaction mechanisms, which are outlined in the following subsection.

2.2.5 Reactions in Electron Ionization

Fragmentation reactions in electron ionization mass spectrometry follow well-characterized patterns that are not random but are reproducible pathways determined by [25] the internal energy distribution and the intrinsic chemistry of the molecular ion. [43, p. 846] Understanding these reaction mechanisms is essential for interpreting EI spectra and for predicting fragmentation patterns from molecular structure. [43, p. 846]

They have been studied extensively by McLafferty and Tureček, see [11]. The following subsections describe some of the most important mechanisms.

Homolytic single-bond cleavage

The most straightforward fragmentation pathway is *simple bond cleavage*, in which a single bond in the molecular ion breaks homolytically, yielding a charged fragment and a neutral radical.

The probability of cleavage at a particular bond depends primarily on bond strength: weaker bonds are more readily broken. [1, p. 30] For aliphatic hydrocarbons, C–C bonds adjacent to branching points or heteroatoms are particularly susceptible to cleavage. [1, p. 258]

Alpha Cleavage

Alpha cleavage is a characteristic fragmentation mode in molecules containing functional groups with heteroatoms. [26, pp. 220–221] In this process, the bond directly adjacent to the functional group is cleaved, producing a resonance-stabilized cation. [26, pp. 220–221] For instance, in ketones and aldehydes, cleavage of the C–C bond next to the carbonyl group generates an acylium ion $\mathrm{R{-}\overset{+}{C}{=}O}$, which is stabilized by the resonance structure $\mathrm{R{-}C{\equiv}\overset{+}{O}}$. [1, p. 230] This reaction pathway often produces intense peaks in the spectrum and serves as a diagnostic indicator for carbonyl-containing compounds. [1, pp. 234–235]

McLafferty Rearrangement

The *McLafferty rearrangement* is a hydrogen-transfer reaction characteristic of carbonyl compounds and other systems with abstractable γ-hydrogens.[26, p. 223] In the commonly depicted mechanism, a six-membered transition state facilitates the transfer of a hydrogen atom from the γ position to the carbonyl oxygen, accompanied by cleavage of the bond between the α and β carbons. [1, p. 265] The result is an enol radical cation and a neutral alkene. [26, p. 224] This rearrangement is particularly valuable for structural elucidation because it reveals connectivity information that would not be apparent from simple cleavage alone. [26, p. 224]

Charge-Induced Fragmentations

Beyond the radical-site-driven cleavages described above, fragmentation can also be initiated by the charge center itself. In such *charge-directed fragmentations*, the positive charge polarizes nearby bonds, facilitating heterolytic cleavage and producing even-electron cations and neutral radicals. [14, pp. 281]

2.3 Computational Mass Spectrometry

Computational tools significantly enhance MS capabilities for structure elucidation and compositional analysis. [15, pp. 1–2] Modern workflows integrate in silico fragmentation models, spectral database searches, and machine-learning methods to accelerate compound identification and enable high-throughput analysis. [25, p. 1] [15, p. 2]

In this section, we outline the key computational strategies in mass spectrometry, including library matching, in silico prediction, and de novo structure inference, before turning to their mathematical and algorithmic foundations.

2.3.1 Library Search

Library search is a spectrum-matching procedure used to identify an unknown compound by comparing its measured spectrum against a reference database of known spectra. [15, p. 8] Each entry in such a database typically stores a reference spectrum together with metadata such as molecular formula, structure, and acquisition conditions.

The comparison operates on two observable quantities: the m/z values of detected peaks and their relative intensities. A query spectrum is considered a match if a reference spectrum reproduces both the positions and the abundance pattern of its peaks sufficiently well.[23, p. 35]

A typical library search workflow proceeds as follows: [15, pp. 1, 5]

(i) **Pre-processing:** the query spectrum is converted into a form suitable for comparison, for example by peak selection, intensity scaling, or related preprocessing steps.

(ii) **Candidate filtering:** database entries may be restricted using precursor mass or precursor m/z information within a tolerance window, reducing the number of candidates that must be evaluated. [5, p. 14]

(iii) **Scoring and ranking:** each candidate reference spectrum is assigned a similarity score against the query, and candidates are ranked accordingly. [15, p. 5]

The most widely used similarity metric is the *cosine similarity*, computed over the vector of peak intensities aligned by m/z: [44, p. 1]

$$\cos(\mathbf{q}, \mathbf{r}) = \frac{\mathbf{q} \cdot \mathbf{r}}{\|\mathbf{q}\| \, \|\mathbf{r}\|}, \tag{2.1}$$

where $\mathbf{q}$ and $\mathbf{r}$ are the intensity vectors of the query and reference spectrum, respectively. A score of 1 indicates a perfect match; lower values indicate increasing dissimilarity.

2.3.2 In Silico Fragmentation

In silico fragmentation predicts how a molecule may dissociate in a mass spectrometer and generates a theoretical spectrum from these simulated fragment ions. [7, p. 1] The standard workflow starts from a candidate molecular structure, applies computational fragmentation rules or learned models to produce plausible fragments, converts these fragments into a predicted spectrum, and finally compares the prediction against the measured spectrum using a similarity score. [7, p. 2] This approach is particularly useful when no exact match is available in spectral libraries, because it enables structure-aware candidate ranking beyond pure reference lookup. [15, pp. 1–2][25, p. 1] At the same time, prediction quality depends strongly on how well the model captures real instrument conditions, ionization-dependent chemistry, and fragment intensities; consequently, performance can degrade for compounds or fragmentation pathways that are underrepresented in the training data or incompletely described by rule sets. [45, p. 1989]

When these in silico predictions remain ambiguous, the workflow may shift to de novo inference directly from spectral constraints.

2.3.3 De Novo Structure Prediction

De novo structure prediction denotes the inference of molecular structure from scratch, i.e., without relying on pre-existing templates or premade reference repositories [46, p. 1]. Importantly, *de novo* characterizes the *template-free* nature of the workflow and does not, by itself, imply a specific modeling paradigm (e.g., physics-based versus data-driven). Accordingly, de novo approaches may incorporate empirical heuristics, statistical learning, or mechanistic constraints, depending on the application. [47, pp. 701–702]

2.3.4 First Principles

In contrast, *first-principles* modeling emphasizes deriving predictions without empirical parameterization, i.e., without fitting to experimental data. [48, p. 1] In physics and chemistry, such first-principles treatments are often realized via *ab initio* electronic-structure methods.

Ab Initio MS Prediction

Ab initio MS prediction refers to computational modeling of mass spectra directly from molecular structure using nonempirical quantum-mechanical methods that are, in principle, independent of experimental fitting [49, p. 1921]. In this sense, ab initio approaches constitute first-principles modeling, because they aim to predict fragmentation behavior and resulting spectra from the underlying physics and chemistry of the molecule rather than from learned correlations. Operationally, ab initio MS prediction typically relies on quantum-chemical calculations (to obtain energetics and feasible fragmentation channels) combined with dynamical or statistical treatments to relate accessible pathways to predicted fragment distributions.

While ab initio methods can provide mechanistic insight and may achieve high accuracy for small molecules, their computational cost often limits scalability to large molecular libraries and complex systems. [47, p. 702]

2.4 Computational Representations of Molecules and Reactions

The encoding of molecular structures and reactions in a machine-processable form is essential for the feasibility of making algorithmic predictions. [50, p. 518]

In computational chemistry, molecules can be represented as graphs. [50, p. 521] Reactions are modeled as graph transformations that break and form bonds. [51, p. 18]

2.4.1 Molecular Graphs

To enable algorithmic processing of molecular structures, we represent them using an abstract graph model that separates topology (which atoms are connected) from optional geometric detail. [51, p. 18]

Figure 2.8 illustrates the abstraction of a molecule as a graph, where vertices represent atoms and edges represent chemical bonds.

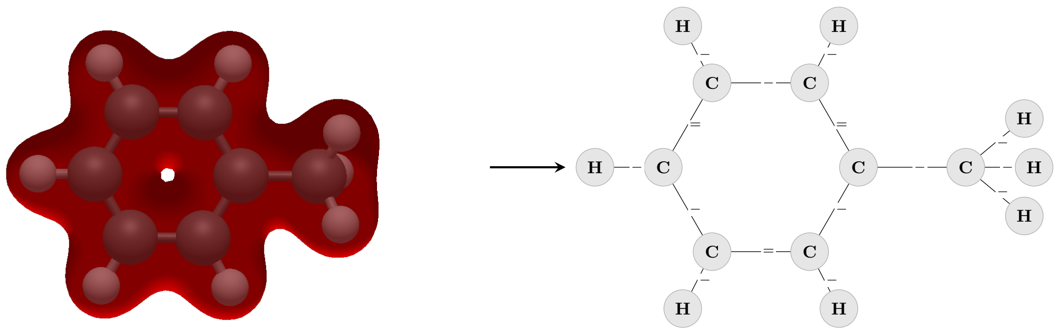


Figure 2.8: Toluene on the left depicted as a quantum-mechanical model with 3D coordinates and electron density, and on the right as a molecular graph abstraction. The graph captures the connectivity of atoms (vertices) and bonds (edges) while abstracting away geometric and electronic details.

This abstraction enables precise operations such as matching and rule-based transformation. [12, pp. 2–3] The formal definition used throughout this thesis is introduced next.

Formal definition

A molecular graph is typically defined as [52, p. 2] [53, p. 3]

$$G = (V, E) \tag{2.2}$$

where:

- V ... Vertices represent atoms

- E ... Edges represent chemical bonds between atoms

In Representations each node may carry attributes such as element type (e.g., C, N, O), formal charge, aromaticity, hybridization state, and isotope label. Likewise, each edge may encode bond order (single, double, or triple), aromatic character, and bond stereochemistry. [54, pp. 4–5] By construction, this basic molecular graph does not include geometric or electronic detail, such as 3D coordinates, molecular conformation, or wavefunction-level information; these require augmented graph models or separate geometric and electronic representations. [54, pp. 3–4] [55, p. 595]

2.4.2 Chemical Graph Transformation Systems

To model chemical reactions algorithmically, molecular graphs are combined with rewrite rules that encode local bond changes. [56, p. 76] In this view, a reaction is not an arbitrary edit of a graph, but a constrained transformation that preserves valid molecular structure while modifying a specific reaction center. [12, p. 2] [57, p. 6]

Double Pushout Graph Rewriting

A standard formalism is the double pushout (DPO) approach. [12, p. 3] A rule is given by

$$p = (L \xleftarrow{l} K \xrightarrow{r} R),$$

where L is the left-hand side pattern to be matched, R is the right-hand side pattern after rewriting, and K is the interface that is preserved. [56, p. 75] Given a match $m : L \to G$ in a host graph G, DPO rewriting constructs an intermediate graph D (deletion pushout complement) [58, p. 680] and then the product graph H (insertion pushout). Intuitively, parts in $L \setminus K$ are removed, parts in $R \setminus K$ are added, and K remains unchanged. [12, p. 3] [59, p. 4]

$$\begin{tikzcd}
L \arrow[d, "m"'] & K \arrow[l, "l"'] \arrow[r, "r"] \arrow[d] & R \arrow[d, "m^*"] \\
G & D \arrow[l, "l^*"] \arrow[r, "r^*"'] & H
\end{tikzcd}$$

Figure 2.9: Schematic of the double pushout (DPO) graph rewriting process. A rule p consists of a left-hand side L, an interface K, and a right-hand side R. Given a match $m : L \to G$ in the host graph G, the DPO approach constructs an intermediate graph D by removing parts of G corresponding to $L \setminus K$, and then constructs the product graph H by adding parts corresponding to $R \setminus K$. The interface K is preserved throughout the transformation, ensuring that the resulting graph H maintains valid molecular structure.

In chemistry, this provides a representation of a reaction center: atoms and bonds in the preserved interface K define shared context, while changes between L and R encode bond breaking/forming events. [60, p. 5]

Derivation Graph

Building on the local rewrite step formalized above, derivation graphs – induced by repeated rule application – provide a global view that shows mechanistically plausible chemical states. [12, pp. 2–3]

A derivation graph, often also called a reaction network, is a higher-level graph that captures all transformation steps generated by a graph rewriting system. [61, Sec. 10] [57, p. 2] Whereas a molecular graph represents the internal structure of a single molecule, a derivation graph represents the *space of possible reactions* between many molecular states. [12, pp. 2–3]

Formally, each node corresponds to an object state (e.g., a molecule, multiset of molecules, or intermediate), and each directed edge corresponds to an application of a rewrite rule that transforms one state into another. [56, p. 75] If a rule p transforms state G into state H, this step is represented as

$$G \xrightarrow{p} H.$$

By collecting many such derivations, one obtains a directed graph whose topology encodes reachable states, alternative pathways, and cycles.

In computational chemistry, derivation graphs are useful for enumerating reaction pathways, identifying branching points, analyzing network connectivity, and ranking plausible mechanistic routes under additional constraints (e.g., energy, rule priorities, or experimental evidence). They therefore provide a natural bridge between rule-based chemistry models and downstream tasks such as pathway search, and machine-learning-based scoring of candidate transformations. [57, p. 2]

These graph-transformation and derivation-graph constructs are implemented in the MØD framework [60], [61], the software used throughout this work to generate fragmentation data; its Python module is referred to as `mod` in the following chapters.

One example derivation graph for the fragmentation rules applied is in Figure 2.10.

Term Mode

Alongside the explicit graph representation, MØD can also handle molecular states in *term mode*, in which structures are expressed as formal terms (tree-like symbolic expressions) rather than as plain graphs. A term is built from function symbols (constructors such as `C`, `O`, or `bond`) applied

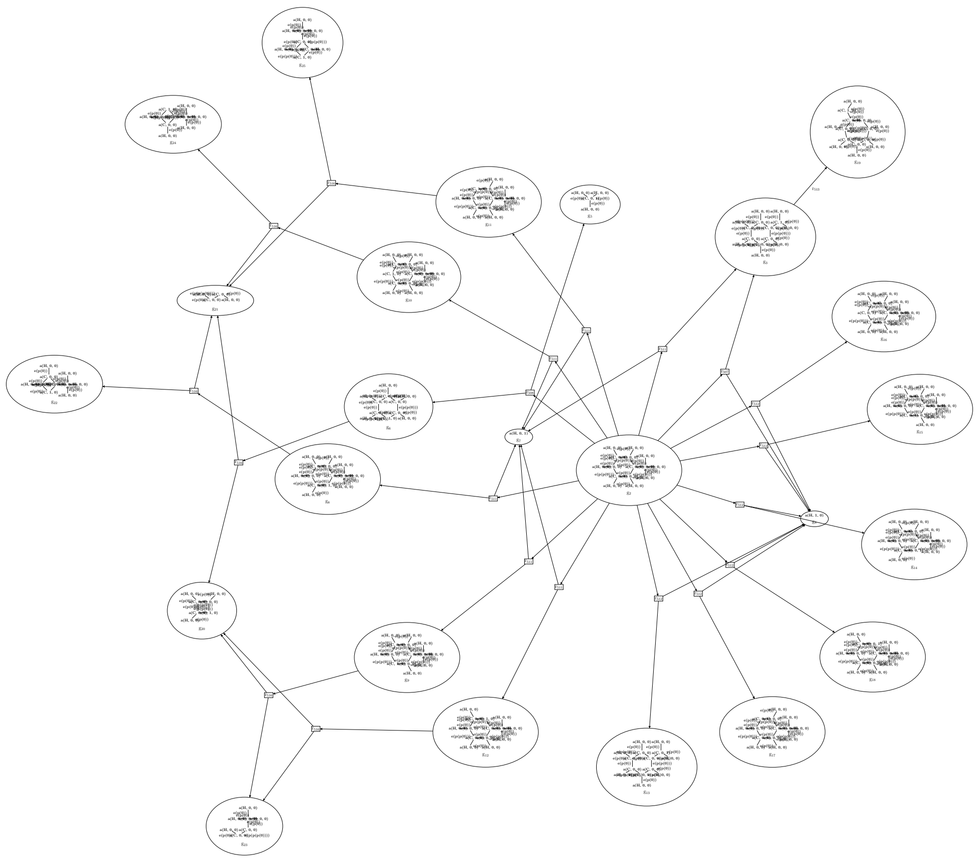


Figure 2.10: Example derivation graph showing reaction pathways for toluene fragmentation. Also illustrates the combinatorial explosion of possible pathways as rules are applied iteratively.

to subterms, much like expressions in term-rewriting systems or functional programming, and is rewritten by the same rule machinery. The resulting object is a *term-graph*: a graph whose attributes carry this symbolic term structure. In this work term-graph inputs are normalised back to plain attributed graphs before encoding (Section 3.2.1), so term mode is relevant here only as the source format of some MØD outputs.

2.4.3 Graph Neural Networks for Molecular Representation Learning

The molecular encoder represents each molecule as a graph whose nodes correspond to atoms and whose edges correspond to chemical bonds. Initial atom features are updated through graph-convolutional message passing, allowing each atom representation to incorporate information from its local molecular environment. The resulting atom embeddings are pooled into a fixed-length graph-level representation, which is subsequently projected into the shared latent space used for comparison with spectral embeddings. This produces a continuous molecular vector that preserves structural information while being suitable for downstream neural-network training. For a visual overview of this workflow, see Figure 2.11.

2.5 Machine Learning in Mass Spectrometry

2.5.1 Problem Formulations

Two main problem formulations are common in the literature:

- **Structure-to-Spectrum Prediction:** Given a molecular structure, predict the corresponding mass spectrum. This is a forward problem that can be approached using mechanistic models, data-driven models, or hybrid approaches that combine both.

- **Spectrum-to-Structure Prediction:** Given a mass spectrum, predict the corresponding molecular structure. As this is an inverse problem, it is often ill-posed, with multiple candidate structures potentially consistent with the same spectrum.

Both tasks are important for different applications: structure-to-spectrum prediction is useful for in silico fragmentation and library generation, while spectrum-to-structure prediction is essential for de novo identification of unknown compounds.

Bidirectional models that can perform both tasks are also possible, and may leverage shared representations to improve performance on both tasks.

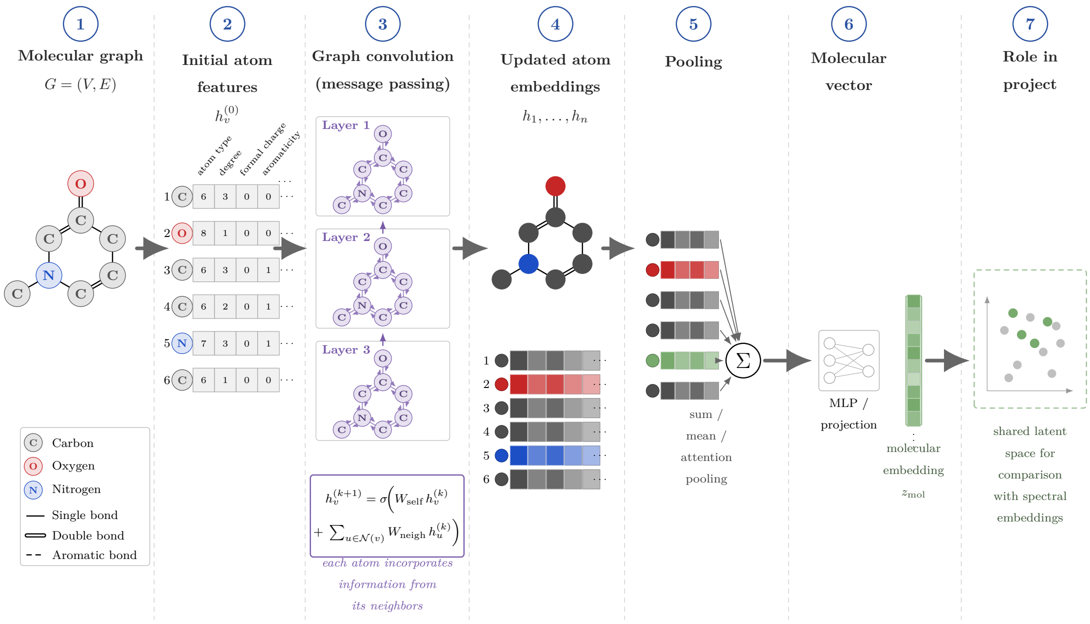


Figure 2.11: Starting from the molecular graph $G = (V, E)$ in **column 1**, where atoms form the nodes and bonds form the edges, each atom is initialized with a feature vector in **column 2** that encodes local chemical properties such as atom type, degree, formal charge, and aromaticity. Through repeated graph-convolution/message-passing layers in **column 3**, these atom representations are updated by aggregating information from neighboring atoms, yielding the context-aware atom embeddings shown in **column 4**. These embeddings are then pooled over all atoms in **column 5** to obtain a fixed-length molecular representation, which is projected in **column 6** into the shared latent space as the molecular embedding z_{mol}. As indicated in **column 7**, this vector can subsequently be compared with spectral embeddings for molecule–spectrum alignment.

2.5.2 Case Study: Generating Molecular Fragmentation Graphs with Autoregressive Neural Networks

A prominent example of integrating physical fragmentation models with neural networks is the work of [8], which proposes an approach to predict tandem mass spectra by directly modeling fragmentation pathways as graphs. Rather than treating spectrum prediction as a black-box prediction task, this method learns to generate fragmentation graphs sequentially: at each step, the model predicts which bonds are most likely to break in the molecular ion, scores the resulting fragments, and iteratively builds a complete fragmentation landscape.

This hybrid approach is particularly relevant to computational MS because it bridges two historically separate paradigms: rule-based in silico fragmentation, which is interpretable but computationally expensive and relies on heuristic intensity prediction, and purely data-driven neural networks, which are fast but lack chemical interpretability. [8, p. 3420] By grounding fragmentation prediction in explicit bond-breaking events, the model maintains mechanistic transparency while leveraging the expressiveness of neural networks for learning complex fragmentation preferences from data.

The method demonstrates several key strengths. Because predictions are generated as explicit fragmentation pathways, the model enables human-interpretable explanations: chemists can inspect why particular fragments appear and evaluate whether predicted breakages are chemically plausible. Also the approach shows particular utility for natural products and complex scaffolds where fragmentation patterns deviate from those captured by simple bond-breaking rules, improving metabolite identification from candidate databases. [8, p. 3427]

However, limitations remain. Prediction quality still depends on the distribution of training data; compounds or fragmentation pathways underrepresented in training tend to be mispredicted, and the model may struggle with instrument-specific quirks or unusual ionization regimes not seen during learning.

2.6 Interpretability in Chemical Machine Learning

2.6.1 Why Interpretability Matters

Interpretability in machine learning refers to the ability to understand and explain why a model makes a particular prediction. [62, p. 1]

For structure elucidation from MS data, an uninterpretable model that ranks candidate structures is of limited value if chemists cannot understand the reasoning. Conversely, models that expose their decision-making process [62, p. 2] —by highlighting which structural features drive spectral predictions or which fragmentation pathways are favored— enable hypothesis generation and accelerate scientific insight.

Furthermore, interpretability serves as a diagnostic tool: when model explanations reveal chemically implausible patterns (e.g., incorrect fragmentation rules or unphysical feature importance), this signals potential model failures or training-data biases that numeric performance metrics alone might not expose.

2.6.2 Classic and General Interpretability Methods

Saliency Maps and Gradient-Based Attribution

Saliency maps compute the gradient of the model output with respect to input features, visualizing regions of input space where small changes produce large changes in prediction. For molecular inputs, saliency can be backpropagated through graph neural networks to identify atoms or bonds whose perturbation most affects the predicted spectrum. [63, p. 4] This approach is computationally efficient and provides local explanations for individual predictions. However, gradients can be noisy, especially near decision boundaries, and may not capture higher-order interactions or saturating effects. [64]

SHAP and Permutation-Based Methods

SHAP (SHapley Additive exPlanations) provides a principled, game-theoretic framework for feature attribution based on Shapley values from cooperative game theory. [65, p. 3] By computing the marginal contribution of each feature when iteratively included or excluded, SHAP assigns consistent importance scores that sum to the difference between the model prediction and a baseline. [65, p. 2] For molecular graphs, SHAP can be applied to atomic features or subgraph patterns, revealing which structural motifs drive predictions. Permutation-based importance methods offer similar insights by repeatedly shuffling features and measuring the resulting drop in model performance. [66, p. 23] These methods are model-agnostic and theoretically grounded, but computationally expensive for large molecules or high-dimensional representations.

Counterfactual Explanations

Counterfactual explanations identify minimal modifications to an input that would change the model output, answering the question: "What would need to change for a different prediction?" [67, p. 4] In the chemical domain, counterfactuals correspond to structural modifications— the loss or gain of functional groups, changes in substitution patterns, or alternative regioisomers— that would yield different predicted spectra. [68, pp. 2–3] These explanations are human-interpretable and actionable, but their discovery is computationally challenging, especially in high-dimensional chemical spaces with complex constraints (e.g., synthetic accessibility, valence rules).

2.6.3 Evaluation of Explanations

The quality of explanations must be evaluated in order to determine whether a method actually achieves its explainability objective and to enable comparison between alternative explanation methods. [69, p. 1] Zhou et al. describe explanation quality in terms of *interpretability* and *fidelity*. Interpretability concerns human understandability and includes the properties of *clarity*, *parsimony*, and *broadness*, whereas fidelity concerns how accurately the explanation reflects model behaviour and includes *completeness* and *soundness*. [69, p. 3]

Based on this view, explanation quality can be assessed through both human-centred and functionality-grounded evaluation. Human-centred evaluation includes application-grounded and human-grounded studies and often considers subjective measures such as trust and confidence together with task-related performance. Functionality-grounded evaluation instead relies on formal proxy metrics tied to specific explainability properties. [69, p. 8] [69, p. 10]

2.6.4 Evaluation Metrics of Prediction Performance

In computational mass spectrometry, evaluation metrics quantify how well predicted spectra match experimental observations. Cosine similarity measures spectral alignment by computing the dot product of normalized peak intensity vectors, accounting for both peak positions and intensities. [70, p. 4] Edit-distance metrics quantify the minimum number of peak modifications needed to transform one spectrum into another, reflecting the structural differences between molecules. [71, pp. 1–2] Wasserstein distances provide a framework for comparing peak distributions, measuring the cost of optimally transporting one spectral peak distribution to another. [72, pp. 1–8] [73] These metrics collectively enable rigorous assessment of model predictions against ground-truth spectra and facilitate comparison of different spectral prediction methods in computational workflows.

Chapter 3

Methodology

3.1 Datasets

This work draws from two main sources of input: measured mass spectra with their corresponding molecular structures as SMILES strings and a set of known fragmentation rules for the ionization method of interest.

3.1.1 Data Sources

Public Databases for spectra and structures

For this work, the National Institute of Standards and Technology (NIST) database for EI GC-MS was selected as the primary data source for the measured mass spectra. NIST is considered the gold standard in the field due to several key advantages: [74, p. 3]

- Comprehensive coverage of compounds
- Standardized data format ensuring consistency across entries
- Spectra experimentally measured under controlled, standardized instrument conditions

Alternative databases such as MassBank, PubChem, HMDB, and MoNa were considered but not utilized in this study.

The raw data from NIST consists of pairs of SMILES strings and their corresponding mass spectra.

Fragmentation Rules

The fragmentation reactions are defined by a set of rules derived from McLafferty's seminal work on mass spectrometry fragmentation [75]. These were implemented as a rule extension to the

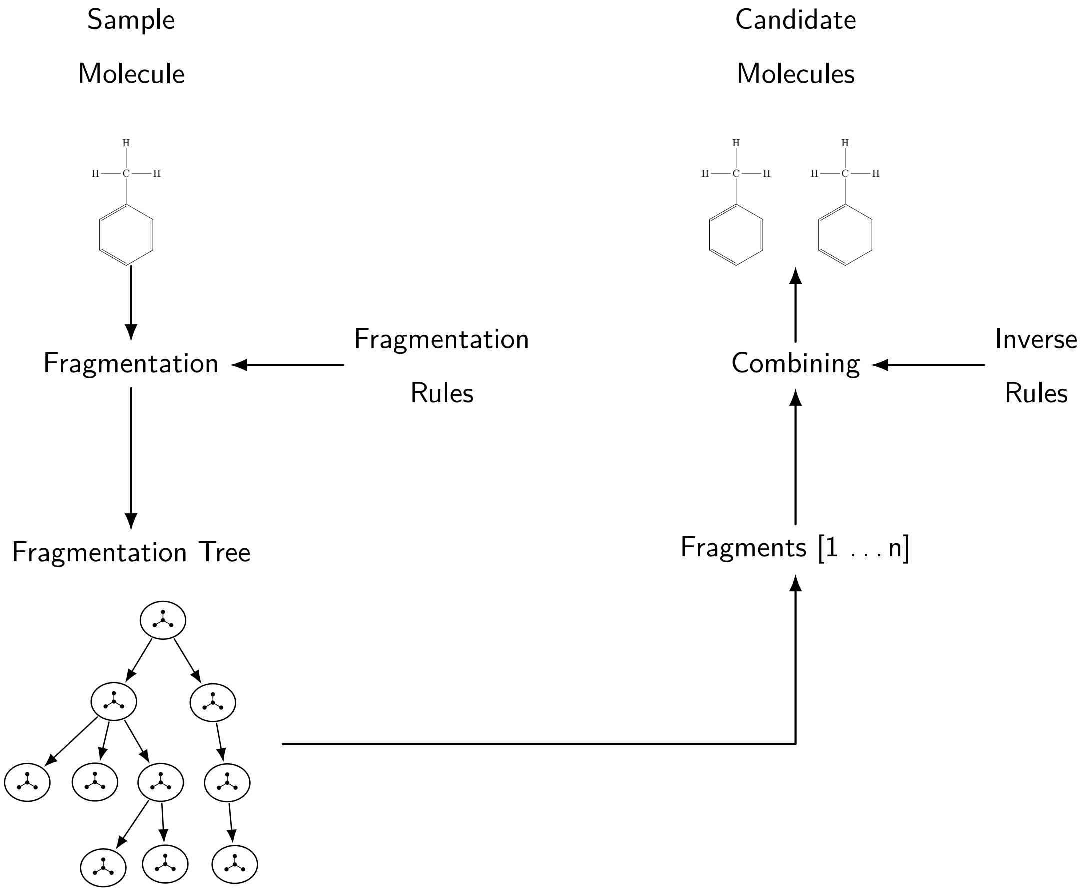


Figure 3.1: Schematic overview of the synthetic data-generation workflow. Starting from a sample molecule, fragmentation rules are applied to generate a fragmentation tree that represents possible fragment formation pathways (left). The resulting fragments can also be recombined according to inverse rules to enumerate candidate molecules consistent with a given fragment set (right). Both directions are produced by the same rule-based graph-transformation pipeline, yielding the molecule–fragment data used to train the model.

MØD framework, enabling the generation of synthetic fragmentation data and the construction of derivation graphs that capture the fragmentation pathways of molecules under study.

3.1.2 Preprocessing - Generating the Synthetic Data

Synthetic training data are generated with `mod` derivation graphs (Figure 3.1). In this representation, nodes correspond to molecular states (fragments), while edges represent individual fragmentation transformations. The resulting graph captures the reaction network and preserves fragment genealogy. This enables the tracing of parent-child relationships and the identification of reachable fragments. Subsequently, this information is utilised to validate predicted fragment assignments against chemically realistic pathways.

The mod strategy framework provides a systematic approach to explore chemical space by guiding fragmentation pathways through both physico-chemical properties and experimental data, such as mass spectra. A key advantage of this framework is its automatic generation of hypergraphs (reaction networks). This automation facilitates the construction of detailed atom flow networks for complex fragmentation patterns, thereby rendering it feasible to track chemical transformations and validate fragmentation predictions on a large scale. [60, p. 7]

Fragmentation Rule Application Strategies

The strategy layer accepts fragmentation rules as input and utilizes them to construct the corresponding reaction network by applying them to molecular structures. Listing 1 shows one representative rule definition used in this workflow.

```
1  IMS_4_9 = mod.Rule.fromDFS(
2      s =
3      "[C]1[C]2[C]3([C]4)[_A+.]5"
4      ">>"
5      "[C.]1" "." "[C]2[C]3([C]4){=}[_A+.]5",
6      name =
7      "radical induced (alpha-)cleavage for a saturated site"
8      " 4.9"
9      " §R1R3R4Y5"
10 )
```

Listing 1: Sample Fragmentation Rule [75, p. 59]

In the mod framework, predicates are callback-based filters evaluated during rule application. More specifically, line 9 of Listing 1 is the critical component where the rule extension defines and enforces chemical constraints, removing candidate sites that violate them during the fragmentation process. This constraint specification forms the core mechanism of the rule extension, implemented as an extension to mod together with the broader application strategy.

Building on this mechanism, the rule extension serves as a *domain-specific language* (DSL) for precisely defining the spatial position and local chemical context of fragmentation sites. Far from a mere convenience, this positional language is essential for encoding complex chemical constraints and enables the systematic specification of reaction sites across molecular structures — without it, encoding such constraints at scale would be infeasible.

On the functional level, the decision flow of this filtering step, enabled by this rule extension, is summarized in Figure 3.2.

To operationalize these chemical constraints in the filtering stage, three predicate checks are

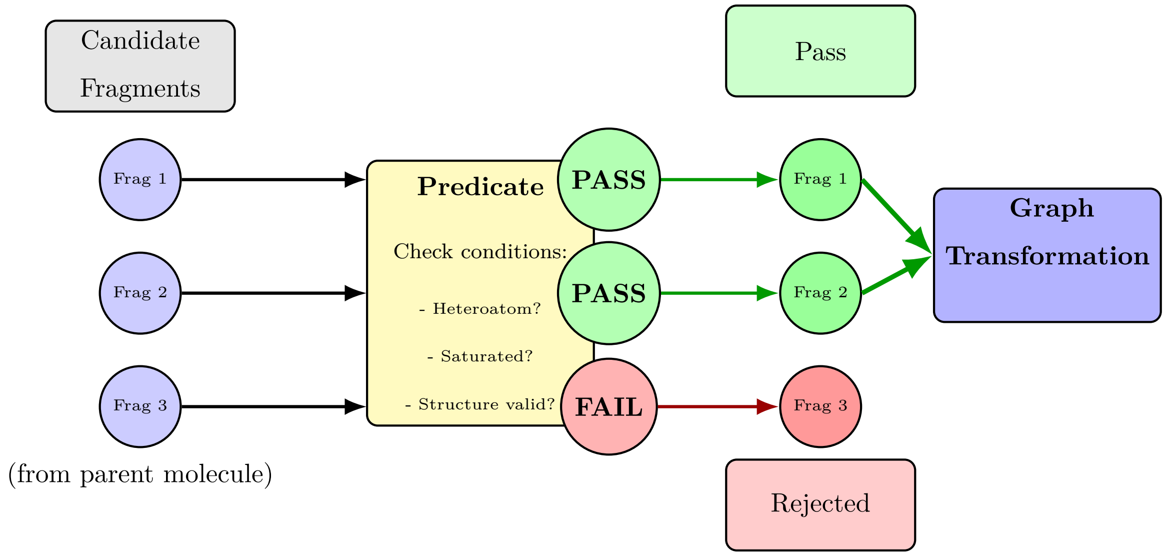


Figure 3.2: Flow diagram explaining predicate filter application at the fragmentation process. During rule application, candidate sites are generated based on the rule's extension pattern. Each candidate is then evaluated by a series of predicate checks (heteroatom, saturation, alkyl)

applied:

Heteroatom Check In order to ensure chemical plausibility during the application of the rule, the heteroatom check retains exclusively those candidate sites which contain at least one non-carbon, non-hydrogen atom within their local neighborhood.

McLafferty's original rule definition [75, p. 85] does not enumerate which elements qualify as heteroatoms. This work therefore adopts the explicit design assumption that the admissible set comprises the organic elements B, C, N, O, P, S, F, Cl, Br, and I, consistent with the typical scope of organic mass spectrometry. This assumption is not enforced programmatically; the check accepts any non-carbon, non-hydrogen atom regardless of element. The resulting limitation is recorded in Section 3.2.5.

The heteroatom check evaluates the neighborhood of the rule-extension location via local graph traversal. A candidate is accepted only if at least one non-carbon, non-hydrogen atom is found, formulated as $\mathrm{BFS} \setminus \{\mathrm{H}, \mathrm{C}\} \neq \varnothing$. If the explored neighborhood contains only carbon and hydrogen, the candidate is rejected. Figure 3.3 contrasts both cases: a valid neighborhood containing a heteroatom and an invalid neighborhood without one.

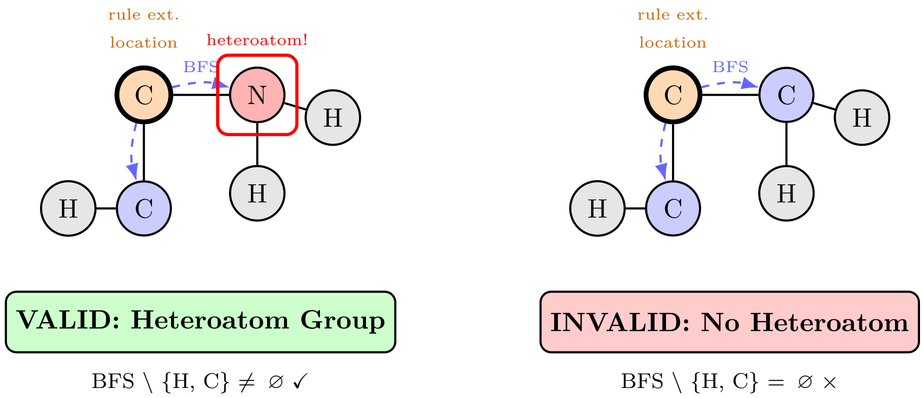


Figure 3.3: Heteroatom check predicate: valid case with detected heteroatom (left) versus invalid C/H-only neighborhood (right).

Saturation Check The saturation check is formulated as a constrained path search between a start and an end atom. A candidate path is accepted only if it follows a carbon backbone through single C–C bonds, while side branches are limited to hydrogen. This removes candidates that include incompatible branch chemistry, such as additional carbon side branches. Figure 3.4 highlights the accepted saturated path in green and marks a rejected carbon branch.

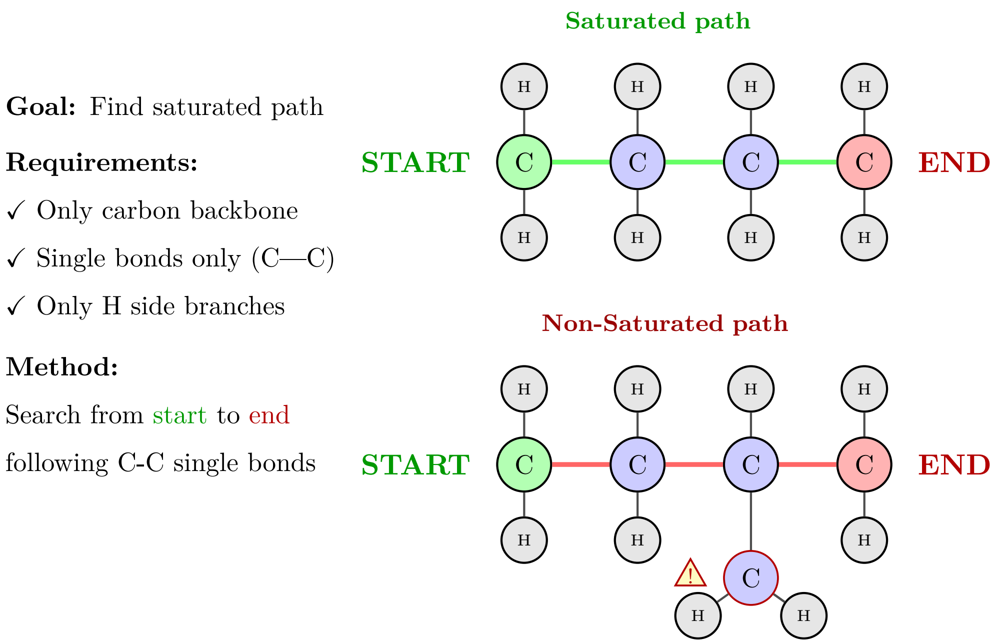


Figure 3.4: Illustration of the concept of a saturated path in a molecular graph. A valid saturated path connects the start and end atoms through a carbon backbone consisting exclusively of single C–C bonds, with only hydrogen atoms as side branches (top). In contrast, a path is not considered saturated if additional carbon side branches are attached to the backbone (bottom).

Alkyl Check The alkyl check ensures that the fragmentation site is part of an alkyl group, restricting fragmentation to aliphatic carbon chains, where such reactions are most likely to occur. This constraint helps filter out fragmentation events in aromatic rings or other structural contexts where different chemical reactivity dominates.

Figure 3.5 visualizes the alkyl check as a breadth-first traversal of the local neighborhood around the rule-extension location. Unlike the saturation check, which validates a *path* between a designated start and end atom, the alkyl check inspects the *surrounding* atoms and accepts the site only if every reachable neighbor is a carbon or hydrogen atom. The presence of any heteroatom—such as the nitrogen marked on the right of Figure 3.5—causes the predicate to fail, since the site no longer lies within a purely aliphatic (alkyl) environment.

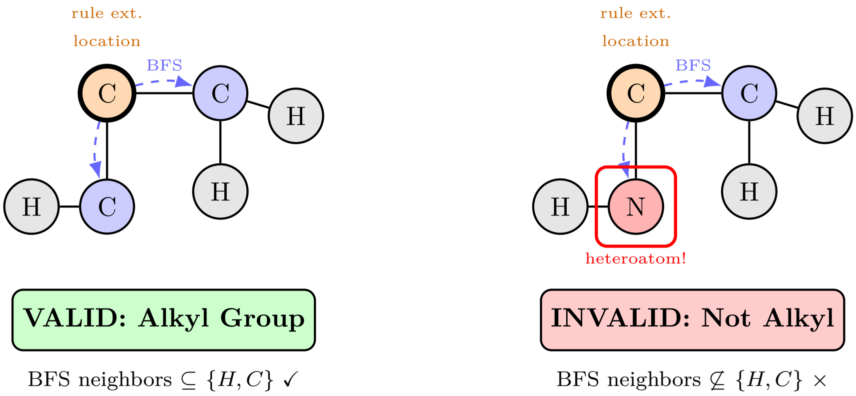


Figure 3.5: Alkyl check predicate formulated as a local-neighborhood traversal. Starting from the rule extension location, a breadth-first search (BFS) is performed over the local molecular graph to verify that all reachable neighboring atoms belong to the set $\{\mathrm{C}, \mathrm{H}\}$. A matched substructure is accepted as an alkyl group if only carbon and hydrogen atoms are encountered (left), whereas the presence of a heteroatom such as nitrogen causes the predicate to fail (right).

3.1.3 Data Splits

The dataset was partitioned in two stages into training, validation, and test subsets. In the first stage, the full set of molecule–spectrum pairs was divided into an 80% training portion and a 20% test portion. In the second stage, the initial training portion was further split into 70% training and 30% validation. This results in a final distribution of 56% training data, 24% validation data, and 20% test data relative to the original dataset.

These ratios were chosen as a pragmatic trade-off between model development and robust evaluation. The 80/20 split keeps a sufficiently large hold-out test set for a meaningful final performance estimate, while still reserving most samples for development.

This splitting strategy serves two purposes. First, it preserves a dedicated hold-out test set that is never used during model development and is therefore reserved for the final evaluation. Second, it provides a separate validation set for model selection, hyperparameter tuning, and intermediate performance monitoring. In this way, the validation set supports iterative model development, while the test set remains an unbiased reference for the final assessment.

The splits are made with `shuffle=False`, so the data are partitioned in their original order rather than being randomly permuted beforehand. The split is therefore deterministic and reproducible by construction; no random permutation is involved, and the `random_state` setting has no effect in this configuration.

An additional technical constraint concerns the fragment representation. The fragment vocabulary is constructed only from the training set and then reused unchanged for the validation and test sets. This ensures that the dimensionality of fragment-based model components remains fixed across all stages of training and evaluation. It also prevents information leakage from validation or test data into the feature space used during training. In practical terms, this means that fragment identifiers available during evaluation are restricted to those observed in the training data, which is consistent with a realistic supervised learning setting.

3.2 Latent Representation Schemes

In this work, we encode molecular fragmentation data through three tightly coupled representation streams:

1. **Molecular states** as chemistry-aware graphs
2. **Derivation dynamics** as a hypergraph over molecular states and transformation rules
3. **Mass spectra** as peak-intensity vectors aligned with structure-derived latent states

The core design principle is a two-level encoding pipeline: first, each molecular state is encoded at the atom and bond level; second, these state-level encodings are used as node features in a derivation hypergraph. This preserves local chemical structure while explicitly representing multi-step fragmentation logic.

Figure 3.6: Overview of the two-level molecular encoding pipeline: local graph encoders capture atom- and bond-level chemistry, while global hypergraph encoders model multi-step fragmentation over embedded molecular states.

In particular, a molecule is not encoded as a SMILES sequence, fingerprint, or direct spectrum-oriented tokenization. Instead, it is represented as an explicit graph object derived from the MØD framework, then mapped into a dense latent embedding, and finally placed into a derivation-hypergraph context together with learned rule representations.

3.2.1 Molecular Graph Encoding

PyTorch Geometric (PyG) [76] is a library that extends PyTorch to enable efficient processing of graph-structured data. It provides specialized data structures and operations optimized for graph neural networks (GNNs).

For molecular graph encoding, each molecular state is converted into a PyG graph object. Atoms are represented as nodes and bonds as edges, allowing GNN architectures to learn from both node features (atomic properties) and edge features (bond information).

Node features are deliberately minimal and chemistry-focused. For each atom, the feature vector contains:

- atom identity (`atom_id`), the atomic number Z (number of protons)
- formal charge (`charge`), the atom's formal charge
- radical state (`radical`), whether the atom carries an unpaired electron

Hence, the node representation is

$$x_i = [\texttt{atom\_id}, \texttt{charge}, \texttt{radical}].$$

Edge features encode bond type as one scalar value, with single, double, triple, and aromatic bonds mapped to 1.0, 2.0, 3.0, and 1.5, respectively. Connectivity is stored in sparse Coordinate (COO) format, where edges are specified as pairs of row and column indices, enabling efficient storage and computation for graphs with varying density.

Before featurization, preprocessing attempts to detect term-graph inputs and converts them into standard graph form where needed. The objective of the normalization step is to ensure the encoder receives a consistently formatted molecular graph representation.

The library handles common challenges in graph processing, such as varying graph sizes and irregular connectivity patterns, through batching mechanisms that stack multiple graphs into a single tensor representation.

PyG's core strength lies in its message-passing framework, which enables efficient computation of graph convolutions. In each layer, nodes aggregate information from their neighbors through learned transformations, allowing the network to capture hierarchical structural patterns relevant to fragmentation behavior. Additionally, PyG provides pre-implemented layers for various GNN architectures (GraphConvolution, GraphAttention, etc.) and inherent permutation invariance, ensuring that the model is robust to different node orderings.

For this work, we utilize PyG's graph convolutional layers to embed molecular structures into a latent space that captures chemical properties while remaining efficient to compute. The flexibility of PyG's API allows seamless integration with standard PyTorch components for downstream prediction tasks.

3.2.2 Two-Level Molecular Latent Representation

After graph featurization, each molecular graph is passed to a learned graph encoder, which compresses it into a dense state embedding $\mathbf{h}_{s_i} \in \mathbb{R}^{d_h}$, one vector per molecular state.

The central architectural distinction is that molecules are not treated as isolated instances. Each molecular state corresponds to a vertex in the *derivation graph*, the functional object that records fragment genealogy. For computation, the derivation graph is realised as a hypergraph data structure, in which a single hyperedge can connect all fragment vertices produced by one transformation step. Throughout this work, "derivation graph" refers to this functional object, whereas "hypergraph" denotes only its concrete data-structure encoding. Encoding each state graph in turn yields one embedding $\mathbf{h}_{s_i}$ per vertex, so that the node features of the derivation graph are not raw atom features but learned embeddings of complete molecular states.

This yields two explicit representational levels:

1. Lower level: each molecule is a molecular graph (atoms and bonds).

2. Higher level: each molecular state becomes a vertex in the derivation graph.

Molecule encoding therefore proceeds in three steps: chemistry-aware graph featurization, graph-to-vector compression into a dense state embedding $\mathbf{h}_{s_i}$, and finally use of this embedding as the vertex feature in the derivation graph.

In the resulting derivation-graph data structure, each vertex carries its state embedding $\mathbf{h}_{s_i}$ as node feature, the incidence structure records which vertices participate in each hyperedge, and every hyperedge is annotated with a learned embedding of the transformation rule (or rule set) that produced the corresponding derivation step (Section 3.2.3).

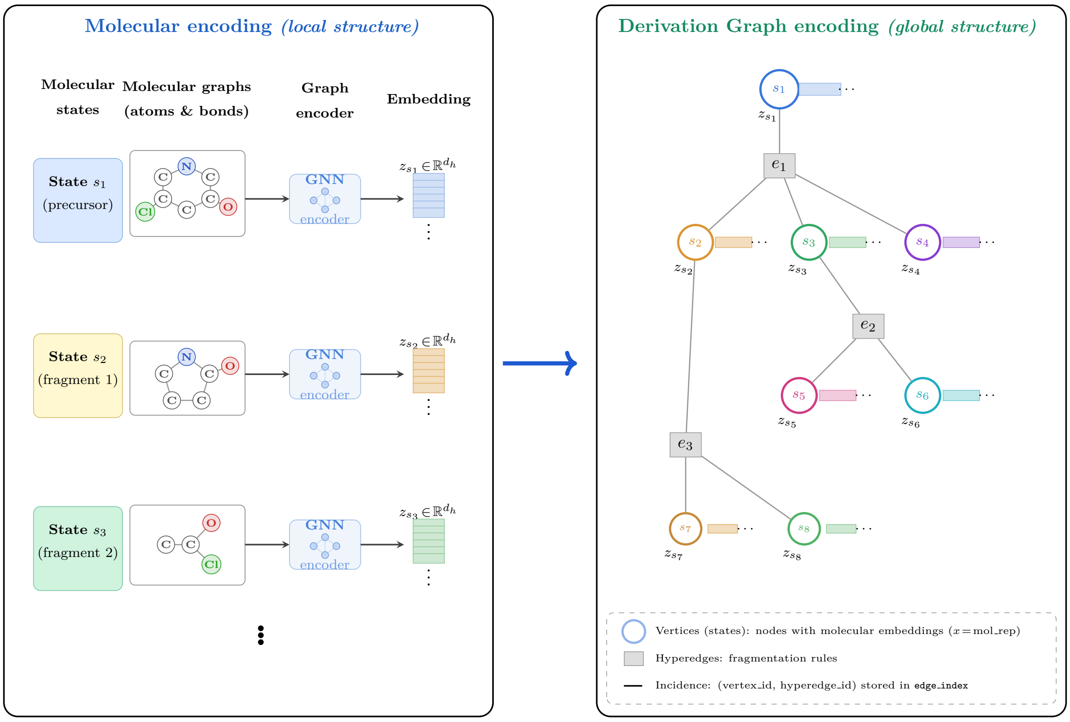


Figure 3.7: In the first stage, each molecular state is represented as an atom-bond graph and encoded by a learned graph encoder into a dense state embedding. In the second stage, these state embeddings are integrated into the derivation graph as vertex features. Thus, vertices correspond to molecular states (whose node feature is the state embedding), while hyperedges and incidence relations encode the higher-level fragmentation structure.

3.2.3 Rule Representation and Hyperedge Annotation

Rule processing is handled separately from molecule processing. Unlike a molecular state, a rule is not expanded into a chemically rich multi-node graph; instead, each rule is represented compactly as a single-node graph whose only feature is its indexed rule identifier.

Rule handling then proceeds in three stages—collection, encoding, and aggregation. If several rules are associated with one hyperedge, their embeddings are mean-pooled into a single hyperedge-level vector. As a result, each hyperedge is annotated with a learned representation of the transformation rule (or rule set) that produced that derivation step.

3.2.4 Spectral Vector and Embedding Representation

Each mass spectrum is treated as a set of tuples $(m/z, I)$, where m/z is the mass-to-charge ratio and I is the corresponding peak intensity. For model input, these tuples are converted into peak-intensity vectors over m/z bins. This tuple-based encoding preserves peak information while producing a dense, differentiable input for neural encoders. These vectors are embedded into a spectral latent space, which is trained to be compatible with the structure-derived latent representation. A shared latent view integrates fragmentation topology, molecular-state embeddings, and observed spectral evidence in downstream learning tasks.

The conversion of a peak list into a latent embedding z_{spec} proceeds in three steps (Algorithm 1). First, the variable-length tuple list is *binned* into a fixed-length vector by discretizing the m/z axis into B equal bins and accumulating peak intensities per bin, yielding an input whose dimension is independent of the number of peaks. Second, an elementwise square root *compresses* the wide intensity range, and L_2 normalization removes overall-scale differences so that the encoder responds to the relative-intensity pattern rather than absolute magnitude. Third, the trained encoder Enc_{spec} maps this vector to $z_{\text{spec}} \in \mathbb{R}^d$, the spectral embedding that the contrastive objective (Equation (3.7)) aligns with the structure-derived representation.

3.2.5 Disadvantages

Despite these representational advantages, the proposed encoding pipeline also introduces several methodological and practical limitations:

- Loss of information due to dimensionality reduction from molecular graphs to fixed-size latent vectors
- Limited interpretability of latent dimensions and pooled rule embeddings
- Potential propagation of preprocessing artifacts into learned representations
- Risk of over-squashing and long-range dependency loss in message-passing encoders
- The set of admissible heteroatoms is not enforced in code but treated as an implicit design assumption (see Section 3.1.2), so structures containing elements outside the assumed organic set are not formally excluded
- The MØD backend used for data generation does not support biradicals, so fragmentation pathways through species with two unpaired electrons cannot be represented

3.3 Model Architecture

This section states the learning problem, then describes the shared three-modality architecture and its forward (structure-to-spectrum) and backward (spectrum-to-structure) branches.

Algorithm 1 Spectrum to latent embedding: from (m/z, intensity) tuples to z_{spec}

Require: peaks: list of (mz, I) tuples; mz_min, mz_max, bin_width; trained Enc_{spec}

Ensure: latent embedding $z_{\text{spec}} \in \mathbb{R}^d$

// **Step 1: bin the sparse peak list into a dense vector**
$B \leftarrow \lfloor(\text{mz\_max} - \text{mz\_min})/\text{bin\_width}\rfloor$ ▷ number of bins
$\text{vec} \leftarrow \mathbf{0}_B$
for all (mz, I) **in** peaks **do**
    $\text{idx} \leftarrow \lfloor(mz - \text{mz\_min})/\text{bin\_width}\rfloor$
    if $0 \leq \text{idx} < B$ **then**
        $\text{vec}[\text{idx}] \leftarrow \text{vec}[\text{idx}] + I$ ▷ accumulate intensity
    end if
end for

// **Step 2: dynamic-range compression + normalization**
$\text{vec} \leftarrow \sqrt{\max(\text{vec}, 0)}$ ▷ elementwise
if $\|\text{vec}\|_2 > 0$ **then**
    $\text{vec} \leftarrow \text{vec}/\|\text{vec}\|_2$ ▷ L2 normalize
end if

// **Step 3: encode into latent space; equivalent to `nn.Sequential`**
$h \leftarrow \text{MLP}_1(\text{vec})$ ▷ "hidden" representation
$z_{\text{spec}} \leftarrow \text{MLP}_2(h)$
return z_{spec}

3.3.1 Problem Formulation

Each training example couples three views of the same molecule:

- a *molecular graph* G representing the precursor structure, with atoms as nodes and bonds as edges;
- a *derivation graph* $D(G)$ whose vertices are fragment states and whose edges encode the fragmentation steps that connect them;
- a measured *mass spectrum*, binned into a fixed-length intensity vector $s \in \mathbb{R}^B$ over B equal m/z bins (Section 3.2.4).

Three encoders map these views into latent vectors,

$$z_{\text{mol}} = \text{Enc}_{\text{mol}}(G), \qquad z_{\text{frag}} = \text{Enc}_{\text{frag}}(D(G)), \qquad z_{\text{spec}} = \text{Enc}_{\text{spec}}(s), \tag{3.1}$$

where the structural encoders produce graph embeddings of dimension d_h, and a set of task heads projects all three views into a shared latent space $\mathbb{R}^d$ in which structural and spectral embeddings can be compared directly.

The model is then trained in two coupled directions:

Forward (structure → spectrum): the molecular and fragment embeddings are fused into a combined latent z_{combined} and decoded into a predicted spectrum $\hat{s} = \text{Dec}_{\text{spec}}(z_{\text{combined}})$.

Backward (spectrum → structure): the spectral embedding z_{spec} is aligned with the molecular embedding z_{mol}, so that at test time the molecules whose embeddings lie closest to a query spectrum can be retrieved.

Both directions are optimised jointly under a single composite objective $\mathcal{L}_{\text{total}}$ (Section 3.3.6); structure-to-spectrum prediction and spectrum-to-structure retrieval therefore share—and mutually constrain—the same latent space. Table 3.1 collects the symbols used throughout this section.

Table 3.1: Notation used in the model description.

Symbol	Meaning
G	molecular graph (atoms as nodes, bonds as edges)
$D(G)$	derivation graph of fragment states for molecule G
$s \in \mathbb{R}^B$	binned mass spectrum over B m/z bins
$\mathbf{h}_{s_i} \in \mathbb{R}^{d_h}$	embedding of an individual molecular state s_i
z_{mol}	molecular (precursor) latent
z_{frag}	fragment / derivation-graph latent
z_{combined}	fused forward latent, $\frac{1}{2}(z_{\text{mol}} + z_{\text{frag}})$
z_{spec}	spectral latent
$\hat{s}$	predicted spectrum
d_h	graph-embedding dimension
d	shared latent (alignment / retrieval) dimension

3.3.2 Overview of the Learning Framework

The methodology presented in this chapter constitutes the central original contribution of this work: to the author's knowledge, it is the first framework to couple MØD-derived fragmentation graphs with a learned two-level molecular representation and to align spectral and structural embeddings in a single bidirectional model. Both the rule-extension DSL and the coupled forward–backward training objective are developed here for the first time.

The implemented learning framework is a bidirectional model that couples molecular structure, fragment-level information, and mass spectra in a shared latent space. As illustrated in Figure 3.8, the model operates on three synchronized inputs, each handled by a dedicated encoder: a molecular graph representing the precursor structure, embedded by a graph neural network; a fragment-derived representation encoding the fragmentation space associated with that molecule, via a derivation-graph backend that encodes fragment graphs and their parent–child relations explicitly; and a binned mass spectrum, represented as a fixed-length vector of binned intensities over the m/z axis and projected into the same latent space through a dedicated spectrum encoder (see Section 3.2.4 for details). These three views are linked through a shared latent representation that supports both forward spectrum prediction and backward structure retrieval, combining a chemically grounded graph representation with a vector-based spectral representation in a single trainable framework.

A central design decision is that the model is not a direct spectrum-to-SMILES generator. Rather, it is a bidirectional embedding model. The forward direction predicts spectra from molecular and fragment information, while the backward direction maps spectra into a latent space in which candidate molecular structures can be retrieved efficiently. This distinction is important for interpreting Figure 3.8: the right-hand side of the figure should not be read as a decoder that explicitly reconstructs a molecular graph, but as a retrieval-oriented inverse path that aligns spectral embeddings with molecular embeddings.

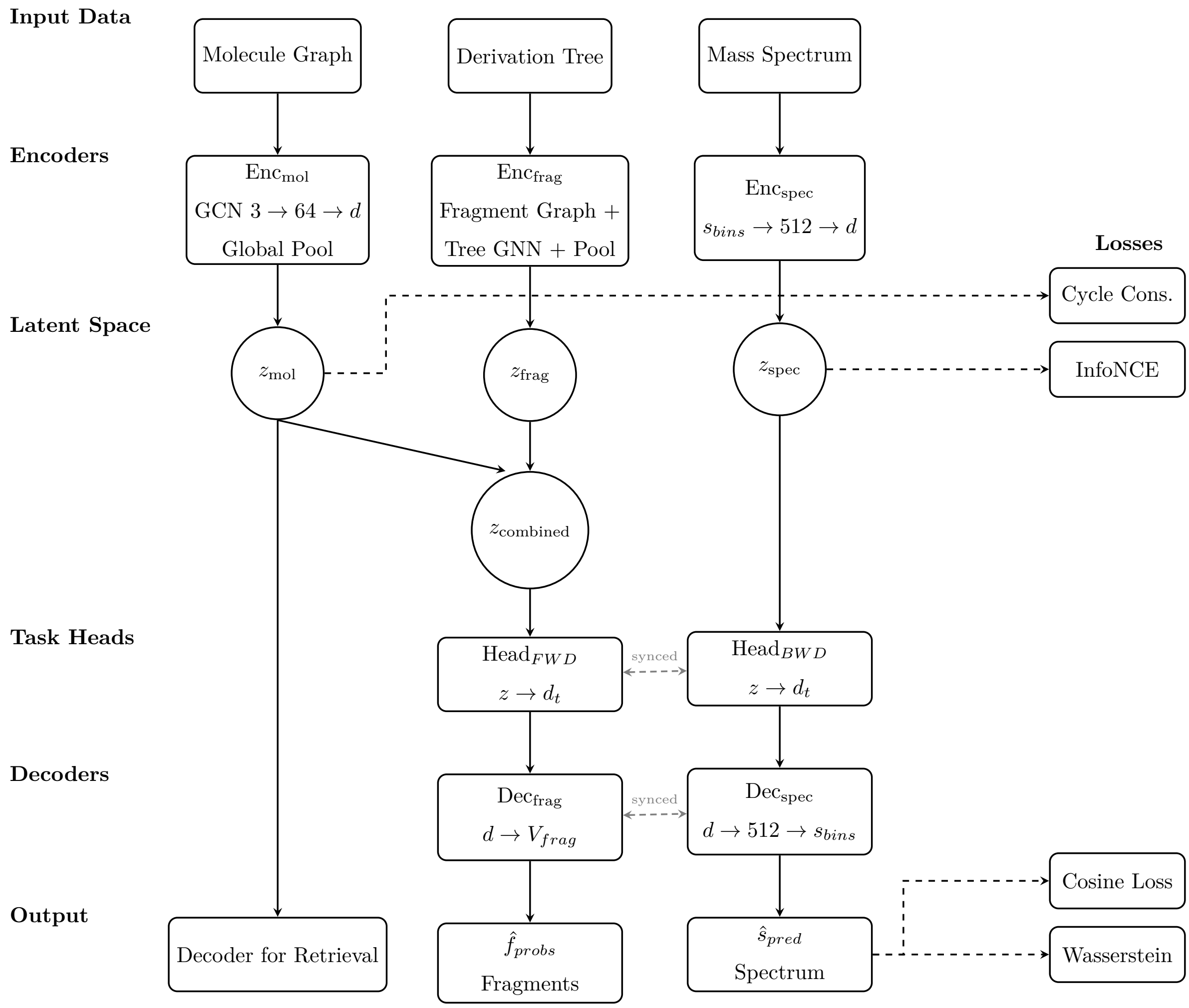


Figure 3.8: Diagram illustrating the overall architecture of the bidirectional translation framework, showing the flow from molecular graph encoding to spectral embedding and back.

3.3.3 Forward Model: Structure-to-Spectrum Prediction

The forward branch corresponds to the left-to-right pathway in Figure 3.8. Its purpose is to predict a mass spectrum from a molecular structure together with fragment-level information derived from precomputed derivation graphs. The molecular encoder $\mathrm{Enc}_{\mathrm{mol}}$ is implemented as a two-layer graph convolutional network. It applies graph convolutions followed by global mean pooling, thereby mapping the input molecular graph to a latent vector z_{mol}. This matches the molecular branch depicted in the figure.

The fragment encoder $\mathrm{Enc}_{\mathrm{frag}}$ is the fragment branch in Figure 3.8 (the *Derivation Tree* input and its encoder) and encodes derivation-graph structure explicitly. For each molecule, the dataset constructs fragment graphs and an explicit derivation graph whose nodes correspond to fragments and whose directed edges encode fragmentation relations. The fragment encoder first embeds the individual fragment graphs using the molecular graph encoder as a backbone. Fragment masses are appended as additional node features, and a second graph neural network is then applied over the derivation graph itself. A global pooling step yields a single latent fragment representation z_{frag}. Accordingly, this branch is implemented as a graph encoder over derivation graphs, not as a simple pooling operation over an unordered fragment bag.

The latent fusion step shown in Figure 3.8 as z_{combined} is implemented by a simple arithmetic average of the molecular and fragment latents,

$$z_{\mathrm{combined}} = \frac{z_{\mathrm{mol}} + z_{\mathrm{frag}}}{2}. \tag{3.2}$$

This fused representation is then passed through the forward task head, $\mathtt{Head}_{\mathrm{FWD}}$, which projects the latent vector into a task-specific subspace.

In latent fusion, the molecular and fragment branches are designed to produce vectors of the same dimensionality, either by sharing the final latent dimension d or by applying linear projection layers in their respective task heads. This ensures z_{mol} and z_{frag} lie in the same vector space and can be combined element-wise. The simple arithmetic average used here Equation (3.2) is a lightweight fusion strategy that treats both modalities equally; more complex fusion (concatenation followed by a projection, attention-based fusion, etc.) could be used but was not required for the presented experiments.

Finally, the shared spectrum decoder $\mathrm{Dec}_{\mathrm{spec}}$ maps this representation to a predicted spectrum $\hat{s}$.

3.3.4 Backward Model: Spectrum-to-Structure Reconstruction

The backward branch corresponds to the spectrum-side pathway in Figure 3.8. Its function is to encode spectra into a latent representation that can be aligned with molecular embeddings and used for later retrieval. The spectrum encoder $\mathrm{Enc}_{\mathrm{spec}}$ receives a fixed-length vector of binned intensities. This encoder is implemented as a regularized stack of residual MLP blocks with normalization and dropout. This makes the spectral branch deeper and more expressive than the simplified notation in the figure might suggest.

Given an input spectrum, $\mathrm{Enc}_{\mathrm{spec}}$ produces a latent vector z_{spec}, which is then passed through the backward task head, $\texttt{Head}_{\mathrm{BWD}}$. This head serves two purposes. First, it provides a task-specific projection for latent alignment with molecular embeddings during training. Second, it defines the representation that is used at test time for inverse retrieval. The projected spectrum embedding is compared to a latent index of molecular embeddings, thus enabling the model to retrieve the most similar candidate structures in embedding space.

At test time this is a spectrum-to-latent-to-candidate pipeline: the model first retrieves candidate molecules by cosine similarity in latent space, then optionally re-ranks them by passing each candidate through the forward branch and comparing its reconstructed spectrum with the query spectrum.

3.3.5 Bidirectionality

The two directions are not learned independently: as shown in Figure 3.8, they share the task heads and the spectrum decoder, and because both are trained jointly within the same architecture, information learned in one direction constrains the other. At test time the molecular encoder builds an index over known candidate structures while the spectrum encoder provides the query, so forward prediction and inverse retrieval become two views of the same embedding problem. The forward model further feeds back into retrieval through reranking, where candidates are rescored by how well their forward-predicted spectra match the observed spectrum.

3.3.6 Loss Function

Metrics

Training a model to predict spectra requires a metric that quantifies how close a predicted spectrum is to its reference. This choice is not merely diagnostic: when the metric is used as the training objective, its geometry determines the loss landscape and therefore what the model is rewarded for reproducing. Two families of comparison are available: *geometric* measures such as the cosine similarity, and *transport* measures such as the Wasserstein distance. The cosine similarity treats each spectrum as a vector and compares two such vectors A and B by the angle θ between them, normalising away their magnitudes so that only their relative direction enters the comparison; Figure 3.9 illustrates this geometry and eq. (3.3) gives the corresponding formula.

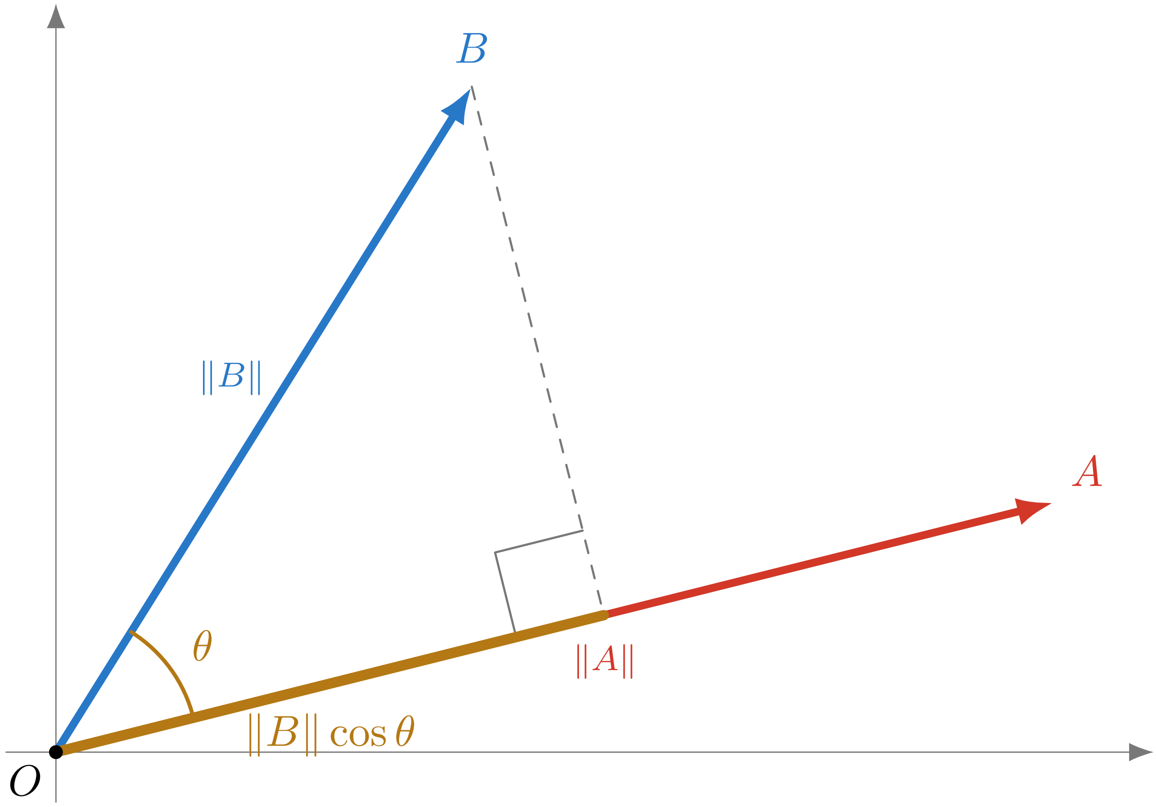


Figure 3.9: Two vectors A and B drawn from a common origin O. The cosine similarity is the cosine of the angle θ between them, $\cos\theta = (A \cdot B)/(\|A\|\,\|B\|)$, and equals the length of the projection of B onto the direction of A (amber) normalised by $\|B\|$. It depends only on the vectors' *directions*: scaling either vector changes its length but leaves θ, and hence the similarity, unchanged.

$$\mathrm{Cos}_{\mathrm{Sim}}(A, B) = \frac{A \cdot B}{\|A\|\,\|B\|} = \frac{\sum_i A_i B_i}{\|A\|\,\|B\|} \tag{3.3}$$

Because the comparison in eq. (3.3) is normalised by both vectors' norms, $\mathrm{Cos}_{\mathrm{Sim}}$ depends only on direction and is invariant to the overall scale of either spectrum.

The Wasserstein distance, also known as the Earth Mover's Distance (EMD), measures the minimum cost of transporting one distribution onto the other, where cost is mass multiplied by the distance it is moved. For two equal-length vectors this reduces to sorting both and summing the absolute differences of matched ranks, a quantity that admits a simple geometric reading. Figure 3.10 draws each sorted vector as a cumulative staircase and shades the horizontal gap between them at every rank; the distance is the total shaded area, normalised by the number of elements.

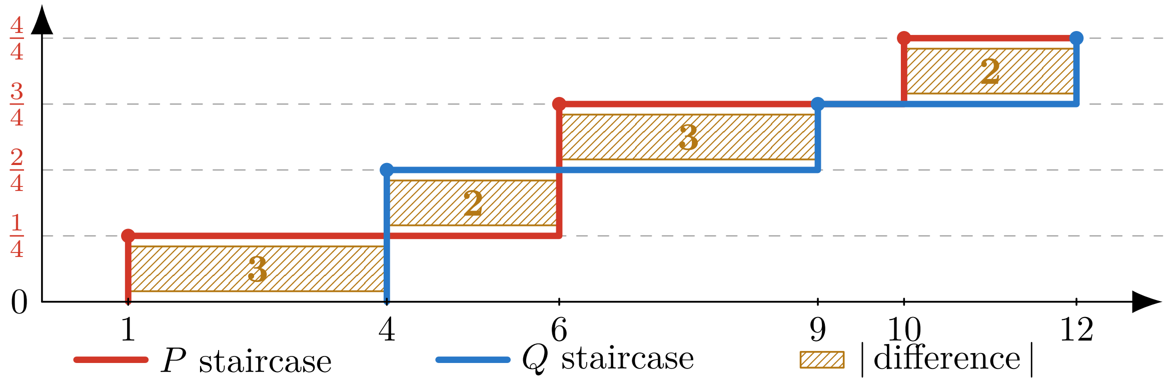


Figure 3.10: Two equal-length vectors, sorted $\tilde{P} = [1, 6, 6, 10]$ (red) and $\tilde{Q} = [4, 4, 9, 12]$ (blue), are drawn as cumulative staircases: the horizontal axis is the value and the vertical axis is the rank index i/n, so each curve steps up by $1/n$ at every element. At each level the hatched amber rectangle spans the horizontal gap between the two staircases, whose width equals the matched-pair difference $|\tilde{P}_i - \tilde{Q}_i|$. The Wasserstein distance is the total hatched area divided by n (see eq. (3.4)), illustrating that the distance accumulates the cost of horizontally transporting one sorted distribution onto the other.

$$\mathrm{WD}(P, Q) = \frac{1}{n} \sum_{i=1}^{n} |\tilde{P}_i - \tilde{Q}_i| = \frac{3 + 2 + 3 + 2}{4} = 2.5 \tag{3.4}$$

The matched-pair form in eq. (3.4) presupposes that both vectors contain the same number of elements, since each rank in $\tilde{P}$ must be paired with a corresponding rank in $\tilde{Q}$. Real spectra rarely satisfy this: two spectra typically differ in the number of detected peaks. To recover a matched-pair comparison without discarding or interpolating peaks, each vector is upsampled to a common length equal to the least common multiple of the two cardinalities, repeating every element enough times to preserve its relative share of the total mass. Figure 3.11 shows this construction for vectors of length four and three.

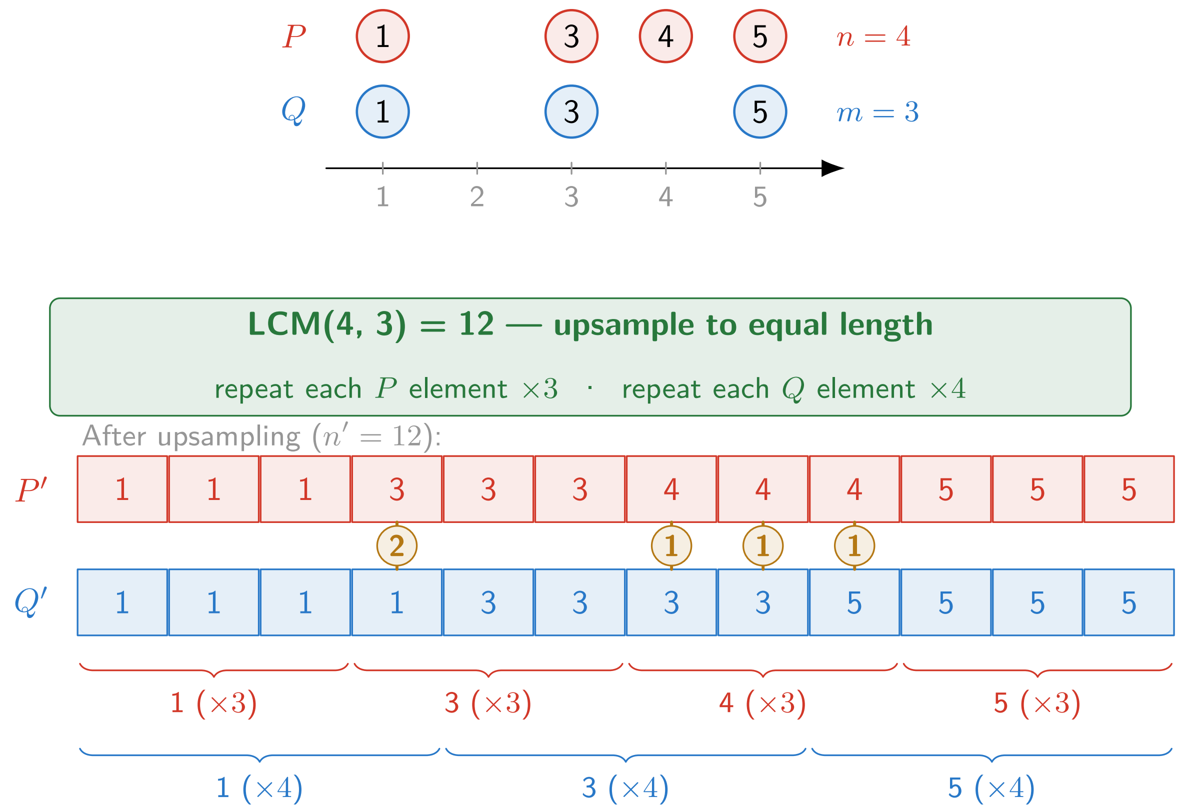


Figure 3.11: The inputs have different lengths, $P = [1, 3, 4, 5]$ $(n = 4)$ and $Q = [1, 3, 5]$ $(m = 3)$, so they are first brought to a common length before matched-pair differences can be formed. Each is upsampled to $\mathrm{LCM}(4, 3) = 12$ by repeating every element of P three times and every element of Q four times (red and blue braces), giving aligned vectors $P' = [1, 1, 1,\ 3, 3, 3,\ 4, 4, 4,\ 5, 5, 5]$ and $Q' = [1, 1, 1, 1,\ 3, 3, 3, 3,\ 5, 5, 5, 5]$. The amber badges mark the four index positions where the upsampled entries differ, with per-index gaps $2, 1, 1, 1$. The Wasserstein distance is the mean of these absolute differences (see eq. (3.5)), showing how repetition lets each value contribute mass in proportion to its share of the original vector, so distributions of different cardinality can be compared directly.

$$P' = [1, 1, 1,\ 3, 3, 3,\ 4, 4, 4,\ 5, 5, 5]$$
$$Q' = [1, 1, 1, 1,\ 3, 3, 3, 3,\ 5, 5, 5, 5]$$

$$
\begin{aligned}
\mathrm{WD} &= \tfrac{1}{12} \cdot (0+0+0+2+0+0+1+1+1+0+0+0) \\
&= \tfrac{5}{12} \\
&\approx 0.417
\end{aligned}
\tag{3.5}
$$

With the Wasserstein distance defined for spectra of arbitrary length, it remains to justify it against the cosine similarity, the conventional measure of spectral agreement. The two differ in what they consider when two peaks fail to coincide. Because the cosine similarity accumulates intensity only at *shared* coordinates, it is blind to how far apart two peaks are: a peak displaced by a fraction of a Dalton and one displaced across the whole axis are treated alike once they no longer overlap. The Wasserstein distance, by contrast, charges for the displacement itself, so a near miss is scored as nearly correct. Figure 3.12 makes this contrast explicit by sliding and rescaling one spectrum against the other and tracking both measures.

The behaviour summarised in fig. 3.12 motivates combining the two measures rather than choosing between them: a loss built on cosine alone provides a near-zero gradient once peaks cease to overlap, while the Wasserstein term supplies a gradient that varies smoothly with displacement. Wasserstein anchors peaks to the right place along the m/z axis; cosine shapes their relative heights once placed (fig. 3.12d). We combine them into a single reconstruction objective,

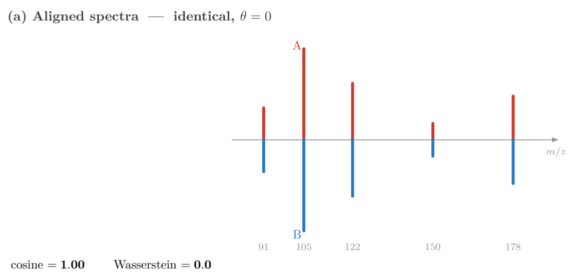
$$
\mathcal{L}_{\text{spec}} = \alpha_{\cos}\,(1 - \cos(P, Q)) + \lambda\,\mathrm{WD}(P, Q), \tag{3.6}
$$

where the cosine similarity is expressed as a dissimilarity $1 - \cos(P, Q)$ so that both terms are minimised at perfect agreement, and the weights $\alpha_{\cos}$ and λ balance intensity-pattern fidelity against positional accuracy. In our configuration $\alpha_{\cos} = 1$ and $\lambda = 0.5$, and during training the cosine term is evaluated on the precursor-mass–masked prediction.

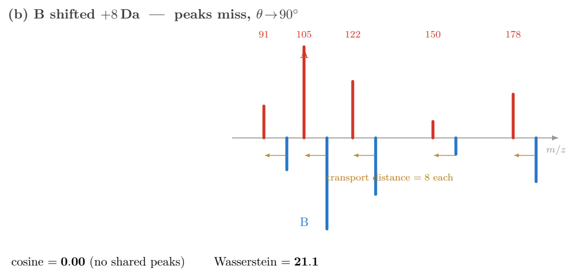
Loss shaping

The training objective is multi-component: a spectral reconstruction term is combined with chemically motivated constraints and a contrastive alignment term. This *loss shaping* stabilises training and ties the shared latent space to chemical structure. Table 3.2 lists every term with the branch in which it is applied and its weight; the two building blocks the table reuses—the InfoNCE alignment objective and the uncertainty weighting—are defined next, and eq. (3.11) assembles the full objective.

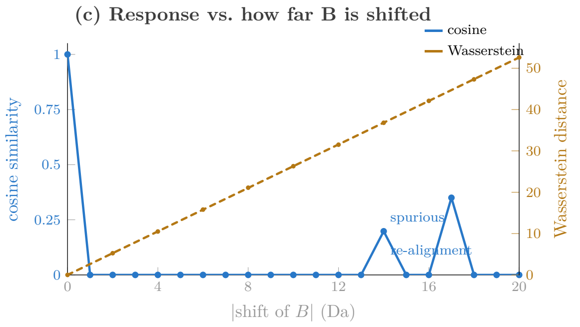
The forward branch is anchored by the reconstruction loss $\mathcal{L}_{\text{spec}}$ (eq. (3.6)) between predicted and true spectra, regularised by a forbidden-region penalty that suppresses intensity beyond the



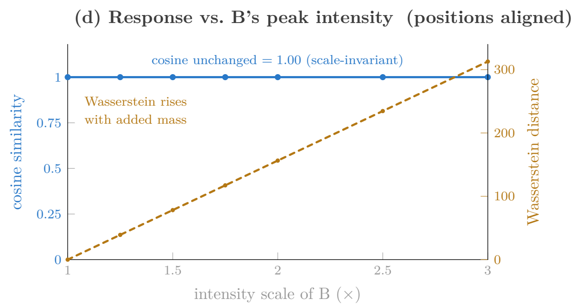
(a) Aligned spectra



(b) B shifted +8 Da



(c) Response vs. positional shift



(d) Response vs. intensity scaling

Figure 3.12: Spectrum A (red) is plotted upward and spectrum B (blue) mirrored downward on a shared m/z axis with non-uniformly spaced peaks.

a Aligned spectra: every B peak meets its A counterpart tip-to-tip, so cosine = 1 (intensity vectors parallel, $\theta = 0$) and Wasserstein = 0 (no mass to move).

b Shifting B by +8 Da slides every peak off its partner; cosine counts only intensity at *shared* positions, so its dot product vanishes and similarity collapses to 0, while Wasserstein measures the work to transport B's mass back onto A (amber arrows) and returns a finite 21.1. c Sweeping the shift, cosine (blue) drops to zero almost at once and stays flat—with *spurious* spikes near 14 and 17 Da where a displaced peak accidentally coincides with a *different* peak—whereas Wasserstein (amber) rises linearly with displacement, degrading gracefully.

d Scaling B's overall intensity with positions held fixed isolates the complementary property: cosine is unchanged at 1.00 because it compares vector *direction*, not magnitude (scale-invariant), while the (un-normalized) Wasserstein distance grows in proportion to the added mass. Together the panels show that the two measures capture different things: cosine rewards matching the *pattern* of relative intensities at fixed coordinates and ignores absolute scale, whereas Wasserstein is sensitive to peak *position* along the axis *and* to total intensity, making it robust to the small calibration shifts that break cosine but responsive to mass differences that cosine discards.

precursor mass, a cycle term that re-encodes the predicted spectrum toward the molecular latent, and a diversity term that discourages collapse to similar predictions within a batch. The backward branch reuses the reconstruction and forbidden-region terms and adds contrastive alignment between spectral and molecular embeddings together with an explicit retrieval-ranking term.

Latent alignment uses the InfoNCE objective, a contrastive loss that encourages the model to bring matching pairs closer together in the latent space while pushing non-matching pairs apart. For a batch of N matched pairs $(z_{\text{spec},i}, z_{\text{mol},i})_{i=1}^{N}$,

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{e^{\frac{\text{sim}(z_{\text{spec},i}, z_{\text{mol},i})}{\tau}}}{\sum_{j=1}^{N} e^{\frac{\text{sim}(z_{\text{spec},i}, z_{\text{mol},j})}{\tau}}}, \tag{3.7}$$

where $\text{sim}(\cdot,\cdot)$ is cosine similarity and τ the temperature. The denominator turns alignment into an N-way classification in which the correct molecule $z_{\text{mol},i}$ must be ranked above the other in-batch embeddings, which serve as negatives without separate mining. The three contrastive terms in Table 3.2 instantiate this form with different embedding pairs.

Selected terms are wrapped in a homoscedastic task-uncertainty weighting,

$$\mathcal{U}(\mathcal{L};\rho) = e^{-\rho}\,\mathcal{L} + \rho, \tag{3.8}$$

where $\rho = \log\sigma^2$ is learned (initialised to 0): $e^{-\rho} = 1/\sigma^2$ down-weights a noisy term while the additive ρ prevents the degenerate solution $\sigma \to \infty$.

Table 3.2: Loss terms with the branch in which each is applied and its weight. The reconstruction components $\mathcal{L}_{\text{cos}}$ and $\mathcal{L}_{\text{wass}}$ together form $\mathcal{L}_{\text{spec}}$ (eq. (3.6)), which is wrapped in the uncertainty term $\mathcal{U}(\cdot;\rho_{\text{spec}})$. Here z_{bwd} is the backward-head projection of z_{spec} and m the hard precursor-mass mask.

Term	Compares	Branch	Weight
$\mathcal{L}_{\text{cos}}$	masked $\hat{s} \odot m$ vs. s (cosine)	both	$\alpha_{\text{cos}} = 1.0$
$\mathcal{L}_{\text{wass}}$	$\hat{s}$ vs. s (1-D EMD over bins)	both	$\lambda = 0.5$
$\mathcal{L}_{\text{forb}}$	intensity beyond precursor mask	both	$\beta = 0.1$
$\mathcal{L}_{\text{cyc}}$	$\text{Enc}_{\text{spec}}(\hat{s})$ vs. z_{mol}	forward	$\lambda_{\text{cyc}} = 0.1$
$\mathcal{L}_{\text{div}}$	off-diagonal batch correlations	forward	$\alpha_{\text{div}} = 0.1$
$\mathcal{L}_{\text{cyc-mol}}$	$\text{Enc}_{\text{spec}}(\hat{s})$ vs. z_{spec}	backward	$\lambda_{\text{cyc}} = 0.1$
$\mathcal{L}_{\text{con}}$	z_{bwd} vs. z_{mol} (InfoNCE)	backward	$\lambda_{\text{con}} = 0.1$
$\mathcal{L}_{\text{cyc-spec}}$	z_{spec} vs. z_{mol} (InfoNCE)	backward	$\lambda_{\text{cyc-spec}} = 0.1$
$\mathcal{L}_{\text{retr}}$	within-batch true-molecule ranking	backward	$\alpha_{\text{retr}} = 1.0$

The model is trained in two phases, each with its own optimiser step. The forward phase minimises

$$\mathcal{L}_{\text{fwd}} = \mathcal{U}(\mathcal{L}_{\text{spec}};\rho_{\text{spec}}) + \beta\,\mathcal{L}_{\text{forb}} + \lambda_{\text{cyc}}\,\mathcal{L}_{\text{cyc}} + \alpha_{\text{div}}\,\mathcal{L}_{\text{div}}, \tag{3.9}$$

and the backward phase minimises

$$\begin{aligned}\mathcal{L}_{\text{bwd}} = {} & \mathcal{U}(\mathcal{L}_{\text{spec}};\rho_{\text{spec}}) + \lambda_{\text{con}}\,\mathcal{U}(\mathcal{L}_{\text{con}};\rho_{\text{con}}) + \beta\,\mathcal{L}_{\text{forb}} \\ & + \lambda_{\text{cyc}}\,\mathcal{L}_{\text{cyc-mol}} + \lambda_{\text{cyc-spec}}\,\mathcal{L}_{\text{cyc-spec}} + \alpha_{\text{retr}}\,\mathcal{L}_{\text{retr}}.\end{aligned} \tag{3.10}$$

Over a full epoch the model therefore optimises the union of these terms,

$$\begin{aligned}\mathcal{L}_{\text{total}} = {} & \mathcal{U}(\mathcal{L}_{\text{spec}};\rho_{\text{spec}}) + \beta\,\mathcal{L}_{\text{forb}} + \lambda_{\text{cyc}}(\mathcal{L}_{\text{cyc}} + \mathcal{L}_{\text{cyc-mol}}) + \lambda_{\text{cyc-spec}}\,\mathcal{L}_{\text{cyc-spec}} \\ & + \lambda_{\text{con}}\,\mathcal{U}(\mathcal{L}_{\text{con}};\rho_{\text{con}}) + \alpha_{\text{retr}}\,\mathcal{L}_{\text{retr}} + \alpha_{\text{div}}\,\mathcal{L}_{\text{div}},\end{aligned} \tag{3.11}$$

with $\alpha_{\text{cos}} = 1.0$, $\lambda = 0.5$, $\beta = 0.1$, $\lambda_{\text{cyc}} = \lambda_{\text{cyc-spec}} = \lambda_{\text{con}} = 0.1$, $\alpha_{\text{div}} = 0.1$, $\alpha_{\text{retr}} = 1.0$, and $\tau = 0.07$; the log-variances ρ_{spec} and ρ_{con} are learned.

Two caveats apply. The reconstruction, forbidden-region, and cycle terms enter *both* phases, so eq. (3.11) is the union of the two phase objectives rather than a single backward pass; and the weighting $\mathcal{U}$ adapts the balance between reconstruction and contrastive terms during training rather than fixing it by hand. The objective is therefore not a simple encoder–decoder reconstruction but a coupled bidirectional system that jointly optimises reconstruction, alignment, ranking, and chemical-plausibility constraints over the shared latent space.

3.3.7 Training Procedure

The two branches are trained *sequentially*, in two separate phases, rather than jointly or alternated per batch. Phase A trains the forward branch (structure→spectrum) to convergence, after which Phase B trains the backward branch (spectrum→structure alignment and spectrum reconstruction); each phase runs for a fixed epoch budget under its own Adam optimiser. Within an epoch, the loss terms of the active branch are summed and back-propagated in a single optimiser step per batch. The spectrum decoder and the task heads—including the learned log-variance weights ρ—are shared across both phases and are therefore updated by both optimisers. The molecule encoder is held as a submodule of the fragment encoder, so it is updated in Phase A as well, even though Phase A's optimiser is nominally constructed over the fragment encoder, the spectrum decoder, and the task heads.

Both phases use identical Adam settings, with no learning-rate schedule, warmup, or gradient clipping. Training is purely epoch-budgeted—there is no early stopping and no validation-based model selection—and the final-epoch weights are retained as the trained model. Accordingly, the validation split is used only *post hoc* for retrieval evaluation (Chapter 4), never for checkpoint selection. Reproducibility rests on a single global seed for initialisation and a fixed split state (Section 3.1.3). The full set of hyperparameters is collected in Table A.1.

Chapter 4

Results and Discussion

4.1 Computation Times of Fragmentation Expansion

Computation times present a significant bottleneck for this approach. The `mod` framework, while powerful in generating comprehensive reaction networks, prioritizes flexibility and completeness over computational efficiency. Combined with the combinatorial explosion of possible fragments and parent molecules, runtime becomes prohibitive as the number of fragmentation rules increases. Since only a subset of known rules were applied in this study, computation times per molecule will rise substantially as additional rules are incorporated, making further optimization necessary to ensure the feasibility of this approach at scale.

4.1.1 Visualization of Computation Times

A regression analysis was performed to model the relationship between molecular complexity and computation time.

Here, molecular complexity refers to Boettcher's complexity measure [77], which summarizes structural intricacy from the molecular graph. It provides a single scalar descriptor that allows molecules of different sizes and connectivities to be compared on a common scale. In this analysis, it is used as a proxy for fragmentation difficulty, although runtime is still influenced by additional factors such as the specific structure of the molecule and the available fragmentation pathways.

The analysis assumes:

- Homoscedasticity of residuals
- Independence of residual observations
- Normality of residual distribution

In Figure 4.1, the regression results are visualized, showing a positive correlation between molecular complexity and computation time, but also show a large variance in computation times for molecules of similar complexity. Note that the threshold of 40 seconds was chosen arbitrarily to exclude trivial expansions.

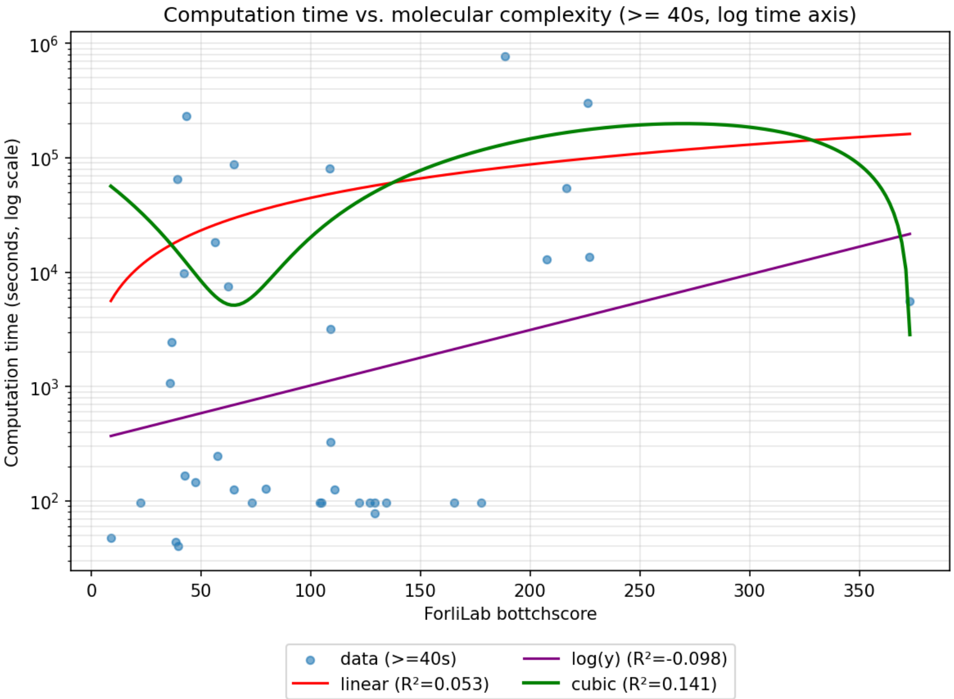


Figure 4.1: Regression of Computation time, excluding molecules that had trivial expansions (time < 40s). The plot shows a positive correlation between molecular complexity and computation time, but also a large variance in computation times for molecules of similar complexity.

So while there is a general trend of increasing computation time with molecular complexity, the large variance suggests that other factors, such as specific structural features or the particular fragmentation pathways available, also play a significant role in determining computation time. This highlights the need for further optimization.

4.2 Evaluation of Fragment Prediction

The evaluation reveals a mixed but informative picture of the proposed framework. On the one hand, the model learns a usable mapping between molecular structures and spectra in both directions. On the other hand, the fragment-aware components do not currently provide a clear empirical advantage over simpler ablated variants. The strongest positive result instead arises from the reranking stage, which substantially improves retrieval quality. To isolate the contribution of individual components, this section includes an *ablation study* comparing the full model against no-fragments, no-derivation-tree, and no-reranking variants. The paired randomization tests underlying this comparison did not yield statistically significant differences for any of the pairwise contrasts, with all p-values exceeding 0.05.

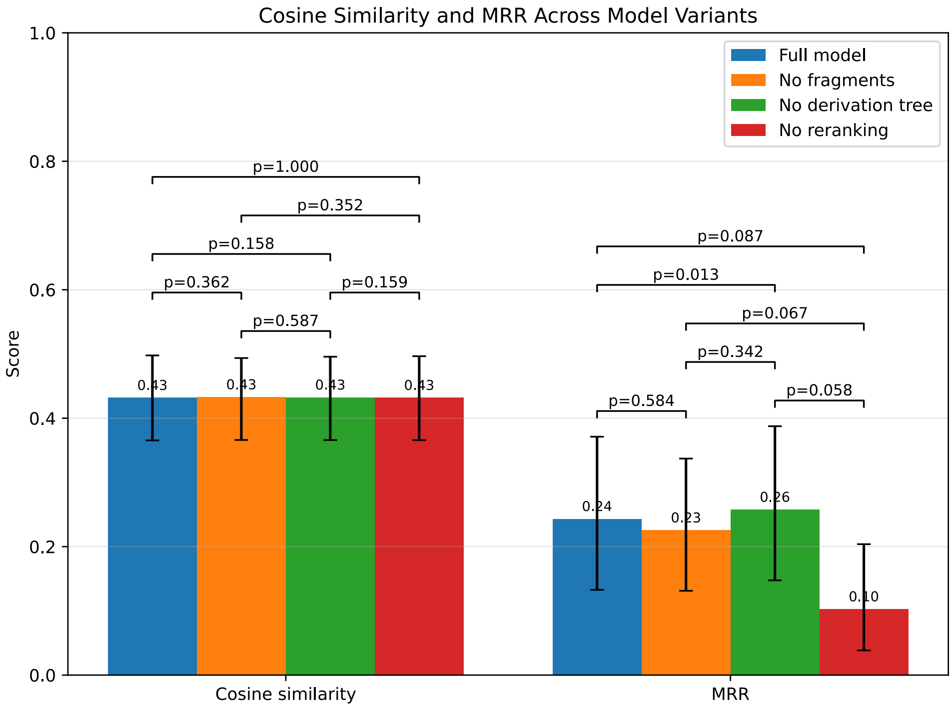


Figure 4.2: Ablation of three pipeline components (fragments, derivation tree, and reranker) against the full model. Cosine similarity measures how similar the query is to the correct target, averaged over all queries; MRR is the mean reciprocal rank of the ground-truth target. Error bars show 95% bootstrap confidence intervals (5000 query resamples). Brackets indicate pairwise p-values from paired randomization tests computed separately for each metric. Cosine similarity remains nearly constant across variants (≈ 0.43), while MRR decreases from 0.24 to 0.10 when reranking is removed. Since all confidence intervals overlap and all pairwise p-values exceed 0.05, these differences should be interpreted as suggestive rather than statistically established.

In this ablation study, with respect to the forward task, all evaluated variants achieve very similar cosine similarity values. The full model reaches a mean cosine similarity of 0.4316, while the no-fragments and no-derivation-tree ablations achieve 0.4327 and 0.4322, respectively. Which is visible in Figure 4.2. These differences are extremely small and indicate that the forward reconstruction quality is largely insensitive to the inclusion or removal of the current fragment-aware branch. In practical terms, the molecular encoder appears to capture most of the predictive signal required for the structure-to-spectrum task, while the fragment representation does not yet provide an additional significant benefit.

The retrieval results are more informative. As shown in Figure 4.3, for the full model, Recall@1 is 0.091, Recall@5 is 0.318, Recall@10 is 0.500, and Recall@20 is 0.864, with a mean reciprocal rank (MRR) of 0.243. This shows that the correct molecule is frequently brought into the top portion of the ranked candidate list, although top-1 identification remains limited. The model therefore behaves more like a candidate narrowing system than a reliable single-shot identifier, which is still useful in practical structure elucidation workflows.

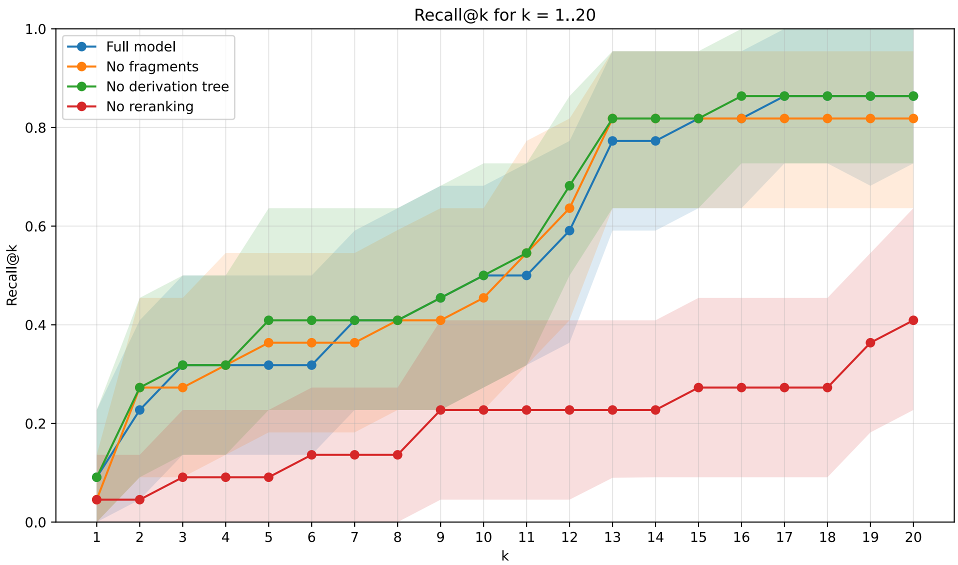


Figure 4.3: Recall@k for $k = 1, \ldots, 20$ across the full retrieval model and three ablated variants. Shaded bands show 95% bootstrap confidence intervals over queries. The full model and the `no fragments` / `no derivation tree` variants cluster together across the full k range, while the `no reranking` variant trails the others at every k, with the largest gap at small k and gradual closure toward $k = 20$.

The clearest effect in the evaluation is observed for the reranking stage. Removing reranking leaves the forward cosine similarity unchanged, as expected, but substantially reduces retrieval quality. Recall@5 drops from 0.318 to 0.091, Recall@10 from 0.500 to 0.227, Recall@20 from 0.864 to 0.409, and MRR from 0.243 to 0.103. This demonstrates that much of the retrieval benefit of the framework arises from combining latent-space retrieval with forward-model-based rescoring. In this sense, the present implementation benefits more from the interaction between the forward and backward branches than from the fragment-aware representation itself.

In contrast, the fragment-related ablations do not support the claim of strong performance. The no-fragments variant demonstrates a marginal improvement in performance when evaluated against the full model in terms of cosine similarity and Recall@5. However, it exhibits a slight decline in performance when assessed against Recall@10, Recall@20, and MRR. In a similar vein, the no-derivation-tree variant has been shown to be highly competitive, with results that are marginally superior to those of the full model on both Recall@5 and MRR. The findings indicate that the fragment branch, in its present state, is not yet demonstrating a substantial and unambiguously favourable signal. It is evident that, at this juncture, the fragment representation is largely redundant given the molecular encoder.

This conclusion should, however, be interpreted cautiously. The test set contains only 22 queries, so recall values change in increments of approximately 0.045. Consequently, differences between variants often reflect only one or two queries. Query-level paired analyses confirm that the differences between the full model and the no-fragments ablation are not statistically convincing for the main metrics, whereas the benefit of reranking is much more substantial. The present evaluation therefore provides stronger support for the usefulness of reranking than for the effectiveness of the fragment-aware branch.

Overall, the results show that the proposed bidirectional framework is functional and that reranking substantially improves retrieval performance. However, they also indicate that the current fragment integration does not yet realize its intended benefit. The main contribution of the present work is therefore not the demonstration of a clear performance gain from fragment augmentation, but rather the establishment of a fragment-augmented bidirectional framework whose strengths, weaknesses, and failure modes can be analyzed systematically. This in turn provides a concrete basis for future work aimed at improving the informativeness and integration of fragment-level intermediate representations.

4.3 Challenges

The thesis is shaped by several interconnected challenges.

First, fragmentation must be represented in a way that is simultaneously chemically meaningful and useful for machine learning.

Second, explicit fragment generation leads to a *combinatorial explosion* of candidate structures and limits the practical scale of the approach.

Third, the absence of unique *ground-truth fragmentation trees* makes fragment-level supervision inherently *ambiguous*.

Fourth, bidirectional learning introduces *competing objectives*, since forward spectrum reconstruction and backward structure retrieval do not necessarily favor the same latent representation.

Finally and **fifth**, the empirical contribution of fragment-aware intermediate representations remains difficult to establish conclusively, especially under *small-sample* conditions and in the presence of strong baseline signals from the molecular encoder alone.

Chapter 5

Future Work

A central limitation of the present framework is that it does not learn fragmentation itself. The model operates on *precomputed* derivation graphs: candidate fragments and their parent–child relations are generated upstream by the rule-based graph-transformation pipeline and are then encoded by the neural model. The learning problem thus begins only *after* a fragmentation space has been constructed. The network learns how to embed, combine, align, and score molecular, fragment, and spectral information, but not which fragmentation events to propose in the first place. This design is appropriate for a first proof of concept, since it keeps the fragment-level representation explicit and the learning problem tractable, but it also marks the clearest boundary of the current approach and thus the most important direction for future work.

5.1 From Fragment Encoding to Fragment Generation

The present implementation is best understood as a *fragment-aware encoder* rather than a full fragmentation simulator. Its fragment branch consumes fragment graphs together with their derivation structure — for example through the derivation-graph backend implemented in `build_derivation_graph_from_collection` and `FragSetEncoderWrapper` — encoding a graph whose nodes are fragment graphs and whose directed edges represent parent–child relations extracted from the precomputed derivation graph. The model therefore receives a structured fragmentation hypothesis as input instead of inferring it autonomously. The natural next step is to move from *encoding* fragmentation structures to *generating* them: rather than assuming the relevant fragment space has already been enumerated, the model would be trained to predict which bonds or substructures are most likely to fragment, in which order, and under which contextual conditions. This would shift the system from symbolic preprocessing with neural downstream learning toward a more integrated neuro-symbolic architecture, considerably closer to a true mechanistic simulator of fragmentation pathways.

5.2 Toward a Generate–Score Architecture

A key source of inspiration for this transition is ICEBERG [10], which separates the task into a *Generate* module that predicts likely fragmentation events and a *Score* module that assigns intensities to the resulting fragments. Crucially, the fragment graph is not treated as fixed input but as a latent object that must itself be predicted: the generative stage is trained to reconstruct an estimated fragmentation directed acyclic graph (DAG), while a second module predicts intensities and hydrogen rearrangements for the generated fragments. The present model already supplies the main ingredients for the *Score* side of such a system. It embeds molecular graphs, fragment-level structures, and spectra into a shared latent space; it predicts spectra from latent representations; and it uses bidirectional training signals such as contrastive alignment and reconstruction losses. What is missing is an upstream module that decides which fragments enter the fragment branch at all. The symbolic derivation graphs produced by the rule-based pipeline could serve as supervision for such a generator — initially as a learned pruning mechanism over the derivation graph, and later as a more ambitious autoregressive model that predicts fragmentation events directly (Figure 5.1).

5.3 Advantages of Learning Fragment Generation

Learning fragmentation generation would offer several advantages over the present setup. First, it would reduce the dependence on exhaustive upstream preprocessing: instead of enumerating a full fragmentation space before learning begins, a learned generator could prioritize only the most informative or probable fragmentation events, thereby lowering the combinatorial burden and improving scalability. Second, it would make the architecture more *end-to-end*. Where the spectrum predictor is currently conditioned on a fragment representation that is treated as given, an integrated design could optimize fragment proposals, fragment embeddings, and spectrum scoring jointly, so the model learns not only how to use fragments but which fragment structures are most helpful for downstream spectrum reconstruction and inverse retrieval. Third, explicit fragmentation generation may improve interpretability. Because the fragment structures are currently externally generated, the model cannot explain why one fragmentation path was preferred over another, whereas a learned generative stage could provide atom- or bond-wise probabilities over fragmentation events and thereby give a more direct account of the model's

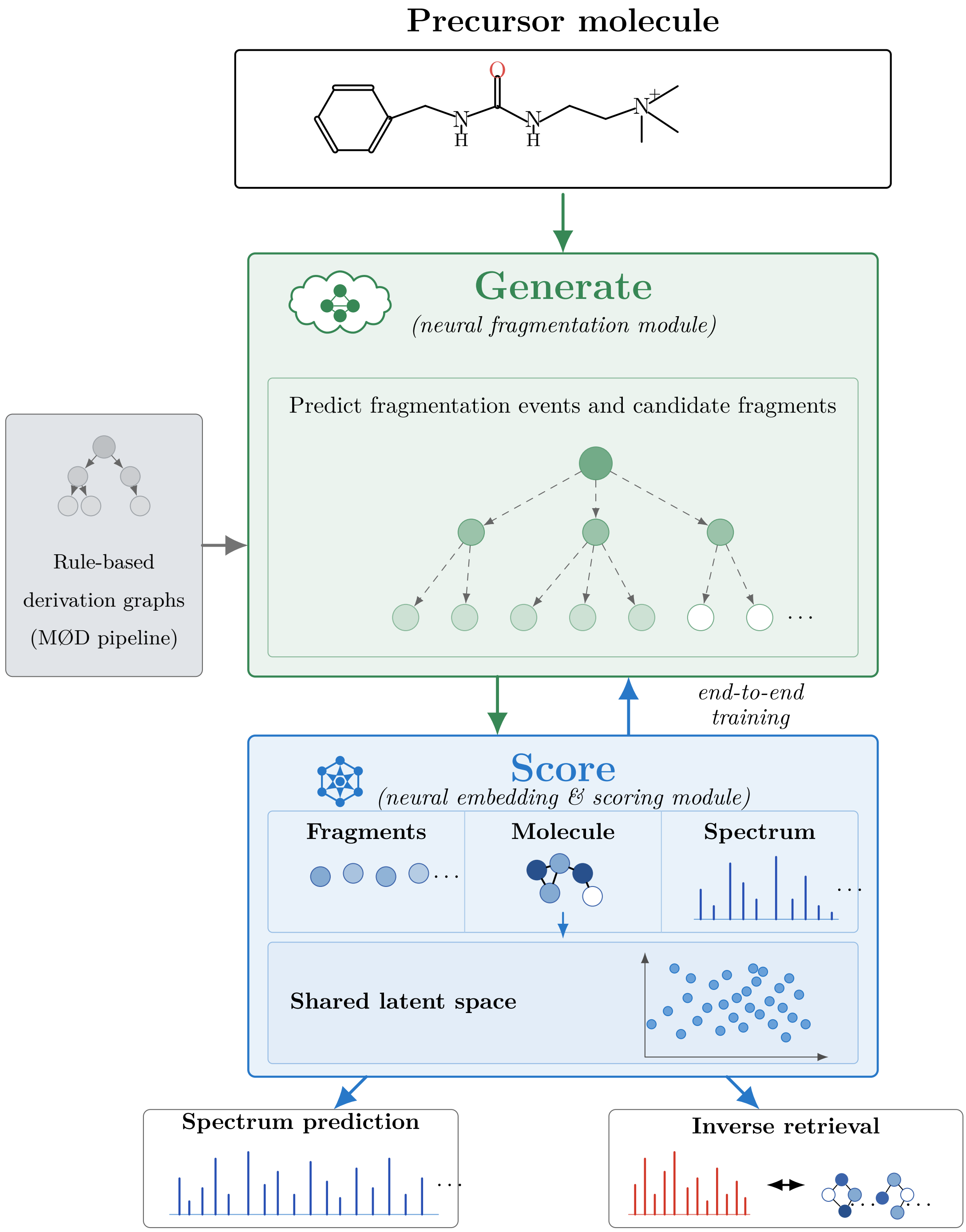


Figure 5.1: A future extension of the present work would be to shift from fragment-structure encoding to fragmentation-process learning. The next major advance will be to let the model learn which fragmentation structures to construct, rather than only how to encode those supplied by an external pipeline.

mechanistic assumptions.

5.4 Scientific and Technical Challenges

This transition is non-trivial and raises several challenges. The first is supervision: spectral libraries generally provide molecule–spectrum pairs, but not ground-truth fragmentation trees. As in ICEBERG, the derivation graphs produced by the rule-based MØD pipeline could serve as proxy labels for a learned generator, but such labels are not guaranteed to be chemically unique or physically exact, and this ambiguity would need to be addressed carefully during training and evaluation. The second is error propagation: once generation becomes learned, mistakes made by the generative stage may propagate into the spectral scoring stage. This argues for staged training, beam search, uncertainty estimation, or latent-variable formulations that can maintain several plausible fragmentation hypotheses rather than committing to a single one. The third is chemical realism: a learned generator should distinguish fragments that are structurally possible from those that are spectrometrically plausible, accounting for effects such as charge retention, neutral loss, and hydrogen rearrangements — again favoring a modular Generate–Score design over a single black-box network.

5.5 A Plausible Development Path

A realistic path forward would proceed in stages. The first stage would keep the current derivation graphs but train a neural module to rank or prune candidate fragmentation steps. The second stage would learn to expand fragmentation trees autoregressively, conditioned on the precursor graph and the partial fragmentation history. The third stage would couple this generator tightly to the existing scoring model so that fragment generation, spectral prediction, and inverse retrieval are optimized jointly. Seen in this way, the present system is not an alternative to Generate–Score approaches but an intermediate step toward them: it demonstrates that fragment-level intermediate representations can strengthen the connection between molecules and spectra, and the next major advance is to let the model learn which fragmentation structures to construct rather than only how to encode those supplied by an external pipeline. This would bring the framework closer to a fully trainable, mechanistically informed, and end-to-end bidirectional model for structure–spectrum translation.

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Appendix A

Source Code Appendix

A.1 Hyperparameter Tables and Configurations

Table A.1 lists the hyperparameters used to train the bidirectional model of Section 3.3.7.

Table A.1: Training hyperparameters.

Setting	Symbol	Value
Optimisation		
Optimiser	—	Adam (one instance per phase)
Learning rate	η	2×10^{-4}
Weight decay	—	1×10^{-4}
Batch size	N	128
Epochs (Phase A + Phase B)	—	$1000 + 1000$
LR schedule / warmup / grad. clip	—	none
Checkpoint selection	—	final epoch
Architecture		
Shared latent dimension	d	128
Graph-embedding dimension	d_h	64
GCN layers	—	2 ($3 \rightarrow 64 \rightarrow 128$)
Spectral MLP width	—	512
Dropout	—	0.1
Normalisation	—	LayerNorm
Spectrum binning		
m/z range	$[mz_{\min}, mz_{\max})$	$[1, 1000)$
Bin width	w_{bin}	1.0 Da
Number of bins	B	999
Loss		
Cosine–Wasserstein weight	λ	0.5 (with $\alpha_{\cos} = 1$)
Contrastive temperature	τ	0.07

A.2 Github Repository

The code for this thesis is available at https://github.com/phireiser/mass-spec

A.3 AI Use Statement

The author used the following AI tools during the writing of this thesis:

- **ChatGPT** (various versions) for language refinement and grammar checking throughout the document.
- **DeepL Write** for improving the style in the document
- **Claude** for structuring and organizing in Chapter 5.
- **GitHub Copilot** for code suggestions and auto-completion in the implementation of the codebase.